

01 GET /alarms

RULES ["compatibility aliases derive only from canonical fields", "source R03 requires alarm_type OOS, rule_code R03_CONSEC, and value null", "source SUMMARY requires rule_code SUMMARY_OOC", "source TRACE requires rule_code TRACE_OOS"]

```
{
  "request": {
    "body": null,
    "header": {},
    "path": {},
    "query": {
      "area": {
        "nullable": true,
        "required": false,
        "schema": {
          "enum": ["ALL", "Etch", "Photo"],
          "type": "string"
        },
        "chamber": {
          "nullable": true,
          "required": false,
          "schema": {
            "type": "string"
          },
          "date_from": {
            "nullable": true,
            "required": false,
            "schema": {
              "format": "date",
              "type": "string"
            },
            "date_to": {
            "nullable": true,
            "required": false,
            "schema": {
              "format": "date",
              "type": "string"
            },
            "equipment": {
            "nullable": true,
            "required": false,
            "schema": {
              "type": "string"
            },
            "include_derived": {
            "nullable": false,
            "required": false,
            "schema": {
              "default": false,
              "type": "boolean"
            },
            "parameter": {
            "nullable": true,
            "required": false,
            "schema": {
              "type": "string"
            },
            "source": {
            "nullable": true,
            "required": true,
            "schema": {
              "enum": ["R03", "SUMMARY", "TRACE"],
              "type": "string"
            },
            "responses": {
              "200": {
                "schema": {
                  "items": {
                    "additional_properties": false,
                    "fields": {
                      "action_code": {
                        "nullable": true,
                        "required": false,
                        "schema": {
                          "enum": ["EQP_HOLD", "MONITORING", "WARNING"],
                          "type": "string"
                        },
                      "alarm_id": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "min_length": 1,
                          "type": "string"
                        },
                      "alarm_type": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "enum": ["OOC", "OOS"],
                          "type": "string"
                        },
                      "area": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "enum": ["Etch", "Photo"],
                          "type": "string"
                        },
                      "chamber": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "min_length": 1,
                          "type": "string"
                        },
                      "chamber_id": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "min_length": 1,
                          "type": "string"
                        },
                      "cl": {
                        "nullable": true,
                        "required": false,
                        "schema": {
                          "type": "number"
                        },
                      "equipment": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "min_length": 1,
                          "type": "string"
                        },
                      "equipment_id": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "min_length": 1,
                          "type": "string"
                        },
                      "fault": {
                        "nullable": true,
                        "required": false,
                        "schema": {
                          "enum": ["FOC", "MFD", "OTH", "RFM", "TMD"],
                          "type": "string"
                        },
                      "lcl": {
                        "nullable": true,
                        "required": false,
                        "schema": {
                          "type": "string"
                        },
                      "lot": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "min_length": 1,
                          "type": "string"
                        },
                      "lot_id": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "min_length": 1,
                          "type": "string"
                        },
                      "mes": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "type": "string"
                        },
                      "mes_status": {
                        "nullable": true,
                        "required": false,
                        "schema": {
                          "enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"],
                          "type": "string"
                        },
                      "notify": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "type": "boolean"
                        },
                      "notify_status": {
                        "nullable": true,
                        "required": false,
                        "schema": {
                          "enum": ["FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"],
                          "type": "string"
                        },
                      "occurred_at": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "format": "date-time",
                          "type": "string"
                        },
                      "parameter": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "min_length": 1,
                          "type": "string"
                        },
                      "parameter_id": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "min_length": 1,
                          "type": "string"
                        },
                      "predicted_fault_code": {
                        "nullable": true,
                        "required": false,
                        "schema": {
                          "enum": ["FOC", "MFD", "OTH", "RFM", "TMD"],
                          "type": "string"
                        },
                      "recipe": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "min_length": 1,
                          "type": "string"
                        },
                      "recipe_id": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "min_length": 1,
                          "type": "string"
                        },
                      "recipe_step_no": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "minimum": 1.0,
                          "type": "integer"
                        },
                      "rule_code": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "enum": ["R03_CONSEC", "SUMMARY_OOC", "TRACE_OOS"],
                          "type": "string"
                        },
                      "seq_no": {
                        "nullable": true,
                        "required": false,
                        "schema": {
                          "minimum": 0.0,
                          "type": "integer"
                        },
                      "source": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "enum": ["R03", "SUMMARY", "TRACE"],
                          "type": "string"
                        },
                      "statistic_type": {
                        "nullable": true,
                        "required": false,
                        "schema": {
                          "type": "string"
                        },
                      "step_no": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "minimum": 1.0,
                          "type": "integer"
                        },
                      "ucl": {
                        "nullable": true,
                        "required": false,
                        "schema": {
                          "type": "number"
                        },
                      "value": {
                        "nullable": true,
                        "required": false,
                        "schema": {
                          "type": "number"
                        },
                      "wafer": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "min_length": 1,
                          "type": "string"
                        },
                      "wafer_id": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "min_length": 1,
                          "type": "string"
                        },
                      "type": "object",
                      "type": "array",
                      "shape": "array",
                      "422": {
                        "schema": {
                          "additional_properties": true,
                          "fields": {
                            "detail": {
                              "nullable": false,
                              "required": false,
                              "schema": {
                                "items": {
                                  "additional_properties": true,
                                  "fields": {
                                    "loc": {
                                      "nullable": false,
                                      "required": true,
                                      "schema": {
                                        "items": {
                                          "type": "union",
                                          "variants": [
                                            {
                                              "type": "string"
                                            },
                                            {
                                              "type": "integer"
                                            }
                                          ],
                                          "type": "array"
                                        },
                                        "msg": {
                                          "nullable": false,
                                          "required": true,
                                          "schema": {
                                            "type": "string"
                                          },
                                          "type": {
                                            "nullable": false,
                                            "required": true,
                                            "schema": {
                                              "type": "string"
                                            },
                                          "type": {
                                            "nullable": false,
                                            "required": true,
                                            "schema": {
                                              "type": "string"
                                            },
                                          "type": "object",
                                          "type": "array",
                                          "shape": "object"
                                        }
                                      }
                                    }
                                  }
                                }
                              }
                            }
                          }
                        }
                      }
                    }
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}
```

02 GET /trace

RULES []

```
{
  "request": {
    "body": null,
    "header": {},
    "path": {},
    "query": {
      "chamber": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        },
      "lot": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        },
      "parameter": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        },
      "wafer": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        }
      }
    },
    "responses": {
      "200": {
        "schema": {
          "items": {
            "additional_properties": false,
            "fields": {
              "measured_at": {
                "nullable": false,
                "required": true,
                "schema": {
                  "format": "date-time",
                  "type": "string"
                },
              "recipe_step_no": {
                "nullable": false,
                "required": true,
                "schema": {
                  "minimum": 1.0,
                  "type": "integer"
                },
              "seq_no": {
                "nullable": false,
                "required": true,
                "schema": {
                  "minimum": 0.0,
                  "type": "integer"
                },
              "value": {
                "nullable": false,
                "required": true,
                "schema": {
                  "type": "number"
                },
              "type": "object",
              "type": "array",
              "shape": "array",
              "422": {
                "schema": {
                  "additional_properties": true,
                  "fields": {
                    "detail": {
                      "nullable": false,
                      "required": false,
                      "schema": {
                        "items": {
                          "additional_properties": true,
                          "fields": {
                            "loc": {
                              "nullable": false,
                              "required": true,
                              "schema": {
                                "items": {
                                  "type": "union",
                                  "variants": [
                                    {
                                      "type": "string"
                                    },
                                    {
                                      "type": "integer"
                                    }
                                  ],
                                  "type": "array"
                                },
                                "msg": {
                                  "nullable": false,
                                  "required": true,
                                  "schema": {
                                    "type": "string"
                                  },
                                  "type": {
                                    "nullable": false,
                                    "required": true,
                                    "schema": {
                                      "type": "string"
                                    },
                                  "type": {
                                    "nullable": false,
                                    "required": true,
                                    "schema": {
                                      "type": "string"
                                    },
                                  "type": "object",
                                  "type": "array",
                                  "shape": "object"
                                }
                              }
                            }
                          }
                        }
                      }
                    }
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}
```

03 GET /parameters

RULES []

```
{
  "request": {
    "body": null,
    "header": {},
    "path": {},
    "query": {},
    "responses": {
      "200": {
        "schema": {
          "item
```

```
s": {"additional_properties": false, "fields": {"LCL": {"nullable": true, "required": false, "schema": {"type": "number"}}, "LSL": {"nullable": true, "required": false, "schema": {"type": "number"}}, "TARGET": {"nullable": false, "required": true, "schema": {"type": "number"}}, "UCL": {"nullable": false, "required": true, "schema": {"type": "number"}}, "USL": {"nullable": false, "required": true, "schema": {"type": "number"}}, "area": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "ctrl_lower": {"nullable": true, "required": false, "schema": {"type": "number"}}, "ctrl_upper": {"nullable": false, "required": true, "schema": {"type": "number"}}, "name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "parameter_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "parameter_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "spec_lower": {"nullable": true, "required": false, "schema": {"type": "number"}}, "spec_upper": {"nullable": false, "required": true, "schema": {"type": "number"}}, "target_value": {"nullable": false, "required": true, "schema": {"type": "number"}}, "unit": {"nullable": true, "required": false, "schema": {"type": "string"}}, "upper_only": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "type": "object", "type": "array", "shape": "array"}}
```

04 POST /documents/search

RULES []

```
{ "request": { "body": {"additional_properties": false, "fields": {"doc_type": {"nullable": true, "required": false, "schema": {"enum": ["MANUAL", "SPEC", "TROUBLESHOOT"], "type": "string"}}, "model_code": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "query": {"nullable": false, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "top_k": {"nullable": false, "required": false, "schema": {"default": 4, "maximum": 10.0, "minimum": 1.0, "type": "integer"}}, "type": "object", "header": {}, "path": {}, "query": {}, "responses": {"200": {"schema": {"items": {"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "content": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "doc_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "model_code": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "score": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": -1.0, "type": "number"}}, "section": {"nullable": true, "required": false, "schema": {"type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object", "array": true, "shape": "array", "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object", "type": "array"}}, "type": "object", "shape": "object"}}
```

05 GET /relations/chambers/{chamber_id}

RULES [{"node_count equals len(nodes)", "relationship_count equals len(relationships)"]

```
{ "request": { "body": null, "header": {}, "path": {"chamber_id": {"nullable": false, "required": true, "schema": {"type": "string"}}, "query": {}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"graph_revision": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "nodes": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"business_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "display_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "label": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "properties": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "type": "object", "type": "array", "relationships": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "target": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object", "type": "array", "root_node_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object", "shape": "object", "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array", "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object", "type": "array", "type": "object", "shape": "object"}}
```

06 GET /agent/runs

RULES []

```
{ "request": { "body": null, "header": {}, "path": {}, "query": {"date_from": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "date_to": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "predicted_fault_code": {"nullable": true, "required": false, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "status": {"nullable": true, "required": false, "schema": {"enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}}, "responses": {"200": {"schema": {"items": {"additional_properties": false, "fields": {"action_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "alarm_source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "approval_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id":
```

```
{
  "nullable": false, "required": true, "schema": {
    "min_length": 1, "type": "string"
  }, "confidence": {
    "nullable": true, "required": true, "schema": {
      "maximum": 1.0, "minimum": 0.0, "type": "number"
    }
  }, "created_at": {
    "nullable": false, "required": true, "schema": {
      "format": "date-time", "type": "string"
    }
  }, "deliveries": {
    "nullable": false, "required": true, "schema": {
      "items": {
        "additional_properties": false, "fields": {
          "channel": {
            "nullable": false, "required": true, "schema": {
              "enum": ["EMAIL", "MES"], "type": "string"
            }
          }, "status": {
            "nullable": false, "required": true, "schema": {
              "enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"
            }
          }
        }, "type": "object", "type": "array"
      }, "fault_code": {
        "nullable": true, "required": true, "schema": {
          "enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"
        }, "fault_color": {
          "nullable": false, "required": true, "schema": {
            "type": "null"
          }, "fault_name": {
            "nullable": false, "required": true, "schema": {
              "type": "null"
            }, "latency_ms": {
              "nullable": false, "required": true, "schema": {
                "minimum": 0.0, "type": "integer"
              }, "llm_model": {
                "nullable": false, "required": true, "schema": {
                  "min_length": 1, "type": "string"
                }, "predicted_fault_code": {
                  "nullable": true, "required": true, "schema": {
                    "enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"
                  }, "recommended_action": {
                    "nullable": true, "required": true, "schema": {
                      "enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"
                    }, "status": {
                      "nullable": false, "required": true, "schema": {
                        "enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"
                      }, "tools": {
                        "nullable": false, "required": true, "schema": {
                          "items": {
                            "additional_properties": false, "fields": {
                              "n": {
                                "nullable": false, "required": true, "schema": {
                                  "min_length": 1, "type": "string"
                                }, "result_summary": {
                                  "nullable": false, "required": true, "schema": {
                                    "min_length": 1, "type": "string"
                                  }, "s": {
                                    "nullable": false, "required": true, "schema": {
                                      "enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"
                                    }, "status": {
                                      "nullable": false, "required": true, "schema": {
                                        "enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"
                                      }, "tool_name": {
                                        "nullable": false, "required": true, "schema": {
                                          "min_length": 1, "type": "string"
                                        }, "type": {
                                          "nullable": false, "required": true, "schema": {
                                            "type": "object", "type": "array"
                                          }, "shape": {
                                            "nullable": false, "required": true, "schema": {
                                              "type": "object", "type": "array"
                                            }, "422": {
                                                "schema": {
                                                  "additional_properties": true, "fields": {
                                                    "detail": {
                                                      "nullable": false, "required": false, "schema": {
                                                        "items": {
                                                          "additional_properties": true, "fields": {
                                                            "loc": {
                                                              "nullable": false, "required": true, "schema": {
                                                                "items": {
                                                                  "type": "union", "variants": {
                                                                    {
                                                  "type": "integer"
                                                }, {
                                                  "type": "array"
                                                }
                                                                    }, "msg": {
                                                                      "nullable": false, "required": true, "schema": {
                                                                        "type": "string"
                                                                      }, "type": {
                                                                        "nullable": false, "required": true, "schema": {
                                                                          "type": "string"
                                                                        }
                                                                      }
                                                                    }
                                                                    }, "type": "object", "type": "array"
                                                                }, "shape": "object"
                                                            }
                                                          }
                                                        }, "type": "array"
                                                    }
                                                  }
                                                }, "type": "object"
                                              }
                                            }
                                          }
                                        }
                                      }
                                    }
                                  }
                                }
                              }
                            }
                          }
                        }
                      }
                    }
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}
```

07 POST /agent/ask

```
RULES ["fault_code and ground_truth_fault_code are forbidden", "metrology.alarm_result is forbidden"]
{
  "request": {
    "body": {
      "additional_properties": false, "fields": {
        "agent_run_id": {
          "nullable": true, "required": false, "schema": {
            "min_length": 1, "type": "string"
          }
        }, "question": {
          "nullable": false, "required": true, "schema": {
            "max_length": 1000, "min_length": 1, "type": "string"
          }
        }, "header": {
          "path": {}, "query": {}, "responses": {
            "200": {
              "schema": {
                "additional_properties": false, "fields": {
                  "answer": {
                    "nullable": false, "required": true, "schema": {
                      "min_length": 1, "type": "string"
                    }
                  }, "confidence": {
                    "nullable": true, "required": true, "schema": {
                      "maximum": 1.0, "minimum": 0.0, "type": "number"
                    }, "evidence": {
                      "nullable": true, "required": true, "schema": {
                        "additional_properties": false, "fields": {
                          "chunk_id": {
                            "nullable": false, "required": true, "schema": {
                              "min_length": 1, "type": "string"
                            }, "doc_id": {
                              "nullable": false, "required": true, "schema": {
                                "min_length": 1, "type": "string"
                              }, "document_id": {
                                "nullable": false, "required": true, "schema": {
                                  "min_length": 1, "type": "string"
                                }, "section": {
                                  "nullable": true, "required": true, "schema": {
                                    "type": "string"
                                  }
                                }, "evidence_items": {
                                  "nullable": false, "required": true, "schema": {
                                    "items": {
                                      "discriminator": "type", "type": "discriminated_union", "variants": {
                                        "AGENT_RUN": {
                                          "additional_properties": false, "fields": {
                                            "excerpt": {
                                              "nullable": false, "required": true, "schema": {
                                                "max_length": 4000, "min_length": 1, "type": "string"
                                              }, "source_id": {
                                                "nullable": false, "required": true, "schema": {
                                                  "min_length": 1, "type": "string"
                                                }, "title": {
                                                  "nullable": false, "required": true, "schema": {
                                                    "min_length": 1, "type": "string"
                                                  }, "type": {
                                                    "nullable": false, "required": true, "schema": {
                                                      "min_length": 1, "type": "string"
                                                    }
                                                  }
                                                }
                                              }
                                            }, "ALARM": {
                                              "additional_properties": false, "fields": {
                                                "excerpt": {
                                                  "nullable": false, "required": true, "schema": {
                                                    "min_length": 1, "type": "string"
                                                  }, "source_id": {
                                                    "nullable": false, "required": true, "schema": {
                                                      "min_length": 1, "type": "string"
                                                    }, "title": {
                                                      "nullable": false, "required": true, "schema": {
                                                        "min_length": 1, "type": "string"
                                                      }, "type": {
                                                        "nullable": false, "required": true, "schema": {
                                                          "min_length": 1, "type": "string"
                                                        }
                                                      }
                                                    }
                                                  }
                                                }, "DOCUMENT": {
                                                  "additional_properties": false, "fields": {
                                                    "chunk_id": {
                                                      "nullable": false, "required": true, "schema": {
                                                        "min_length": 1, "type": "string"
                                                      }, "document_id": {
                                                        "nullable": false, "required": true, "schema": {
                                                          "min_length": 1, "type": "string"
                                                        }, "excerpt": {
                                                          "nullable": false, "required": true, "schema": {
                                                            "min_length": 1, "type": "string"
                                                          }, "section": {
                                                            "nullable": true, "required": true, "schema": {
                                                              "type": "string"
                                                            }, "source_id": {
                                                              "nullable": false, "required": true, "schema": {
                                                                "min_length": 1, "type": "string"
                                                              }, "title": {
                                                                "nullable": false, "required": true, "schema": {
                                                                  "min_length": 1, "type": "string"
                                                                }, "type": {
                                                                  "nullable": false, "required": true, "schema": {
                                                                    "type": "string"
                                                                  }
                                                                }
                                                              }
                                                            }
                                                          }
                                                        }, "GRAPH": {
                                                          "additional_properties": false, "fields": {
                                                            "excerpt": {
                                                              "nullable": false, "required": true, "schema": {
                                                                "min_length": 1, "type": "string"
                                                              }, "graph_revision": {
                                                                "nullable": false, "required": true, "schema": {
                                                                  "min_length": 1, "type": "string"
                                                                }, "relation_id": {
                                                                  "nullable": false, "required": true, "schema": {
                                                                    "min_length": 1, "type": "string"
                                                                  }, "source_id": {
                                                                    "nullable": false, "required": true, "schema": {
                                                                      "min_length": 1, "type": "string"
                                                                    }, "title": {
                                                                      "nullable": false, "required": true, "schema": {
                                                                        "min_length": 1, "type": "string"
                                                                      }, "type": {
                                                                        "nullable": false, "required": true, "schema": {
                                                                          "type": "string"
                                                                        }
                                                                      }
                                                                    }
                                                                  }
                                                                }, "METROLOGY": {
                                                                  "additional_properties": false, "fields": {
                                                                    "excerpt": {
                                                                      "nullable": false, "required": true, "schema": {
                                                                        "min_length": 1, "type": "string"
                                                                      }, "source_id": {
                                                                        "nullable": false, "required": true, "schema": {
                                                                          "min_length": 1, "type": "string"
                                                                        }, "title": {
                                                                          "nullable": false, "required": true, "schema": {
                                                                            "min_length": 1, "type": "string"
                                                                          }, "type": {
                                                                            "nullable": false, "required": true, "schema": {
                                                                              "type": "string"
                                                                              }
                                                                          }
                                                                        }
                                                                      }
                                                                    }
                                                                  }
                                                                }, "TRACE": {
                                                                  "additional_properties": false, "fields": {
                                                                    "excerpt": {
                                                                      "nullable": false, "required": true, "schema": {
                                                                        "min_length": 1, "type": "string"
                                                                      }, "source_id": {
                                                                        "nullable": false, "required": true, "schema": {
                                                                          "min_length": 1, "type": "string"
                                                                        }, "title": {
                                                                          "nullable": false, "required": true, "schema": {
                                                                            "min_length": 1, "type": "string"
                                                                          }, "type": {
                                                                            "nullable": false, "required": true, "schema": {
                                                                              "type": "string"
                                                                              }
                                                                          }
                                                                        }
                                                                      }
                                                                    }
                                                                  }
                                                                }
                                                              }
                                                            }
                                                          }
                                                        }, "type": "array"
                                                    }
                                                  }
                                                }, "limitations": {
                                                  "nullable": false, "required": true, "schema": {
                                                    "items": {
                                                      "type": "str"
                                                    }
                                                  }
                                                }
                                              }
                                            }
                                          }
                                        }
                                      }
                                    }
                                  }
                                }
                              }
                            }
                          }
                        }
                      }
                    }
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}
```

```
ing"}, "type": "array"}}, "predicted_fault_code": {"nullable": true, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "recommended_action": {"nullable": true, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "tools": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result_summary": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "tool_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}}, "type": "array"}}, "type": "object"}, "shape": "object"}, "404": {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"enum": ["APPROVAL_ALREADY_DECIDED", "DEPENDENCY_NOT_READY", "IDEMPOTENCY_CONFLICT", "INCIDENT_ALREADY_PROCESSED", "INCIDENT_ALREADY_RUNNING", "INTERNAL_ERROR", "LEGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_NOT_FOUND", "UNAUTHORIZED", "VALIDATION_ERROR"], "type": "string"}}, "details": {"nullable": false, "required": false, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "message": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "shape": "object"}, "422": {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"enum": ["APPROVAL_ALREADY_DECIDED", "DEPENDENCY_NOT_READY", "IDEMPOTENCY_CONFLICT", "INCIDENT_ALREADY_PROCESSED", "INCIDENT_ALREADY_RUNNING", "INTERNAL_ERROR", "LEGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_NOT_FOUND", "UNAUTHORIZED", "VALIDATION_ERROR"], "type": "string"}}, "details": {"nullable": false, "required": false, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "message": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "shape": "object"}, "503": {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"enum": ["APPROVAL_ALREADY_DECIDED", "DEPENDENCY_NOT_READY", "IDEMPOTENCY_CONFLICT", "INCIDENT_ALREADY_PROCESSED", "INCIDENT_ALREADY_RUNNING", "INTERNAL_ERROR", "LEGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_NOT_FOUND", "UNAUTHORIZED", "VALIDATION_ERROR"], "type": "string"}}, "details": {"nullable": false, "required": false, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "message": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "shape": "object"}}}
```

08 GET /approvals

RULES ["APPROVED or REJECTED requires decided_by and decided_at", "PENDING requires all decision fields null", "only EQP_HOLD actions have approvals"]

```
{ "request": { "body": null, "header": {}, "path": {}, "query": {} }, "responses": { "200": { "schema": { "items": { "additional_properties": false, "fields": { "action_code": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approved_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "approved_by": {"nullable": true, "required": true, "schema": {"type": "string"}}, "chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_by": {"nullable": true, "required": true, "schema": {"type": "string"}}, "decision_comment": {"nullable": true, "required": true, "schema": {"type": "string"}}, "equipment": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "lot": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "predicted_fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "reason": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "type": "object"}, "type": "array", "shape": "array" } }
```

09 POST /approvals/{approval_id}/decision

RULES ["APPROVED or REJECTED requires decided_by and decided_at", "PENDING requires all decision fields null", "only EQP_HOLD actions have approvals"]

```
{ "request": { "body": { "additional_properties": false, "fields": { "decided_by": {"nullable": false, "required": true, "schema": {"max_length": 40, "min_length": 1, "type": "string"}}, "decision": {"nullable": false, "required": true, "schema": {"enum": ["APPROVED", "REJECTED"], "type": "string"}}, "decision_comment": {"nullable": true, "required": false, "schema": {"max_length": 1000, "type": "string"}}, "type": "object"}, "header": {}, "path": { "approval_id": {"nullable": false, "required": true, "schema": {"type": "string"}}, "query": {} }, "responses": { "200": { "schema": { "additional_properties": false, "fields": { "action_code": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approved_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "approved_by": {"nullable": true, "required": true, "schema": {"type": "string"}}, "chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_by": {"nullable": true, "required": true, "schema": {"type": "string"}}, "decision_comment": {"nullable": true, "required": false, "schema": {"max_length": 1000, "type": "string"}}, "equipment": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "lot": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "predicted_fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "reason": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "type": "object"}, "type": "array", "shape": "array" } }
```

```
: true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_by": {"nullable":
: true, "required": true, "schema": {"type": "string"}}, "decision_comment": {"nullable": true, "required
": true, "schema": {"type": "string"}}, "equipment": {"nullable": false, "required": true, "schema": {"mi
n_length": 1, "type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_l
ength": 1, "type": "string"}}, "fault_code": {"nullable": false, "required": true, "schema": {"enum": ["F
OC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "lot": {"nullable": false, "required": true, "schem
a": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min
_length": 1, "type": "string"}}, "predicted_fault_code": {"nullable": false, "required": true, "schema":
{"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "reason": {"nullable": false, "required
": true, "schema": {"min_length": 1, "type": "string"}}, "status": {"nullable": false, "required": true,
"schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "type": "object"}, "shape":
"object"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "re
quired": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false,
"required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integ
er"}]}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "t
ype": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "a
rray"}}, "type": "object"}, "shape": "object"}}
```

10 GET /audit-logs

RULES []

```
{"request": {"body": null, "header": {}, "path": {}, "query": {"actor_type": {"nullable": true, "required
": false, "schema": {"type": "string"}}, "date_from": {"nullable": true, "required": false, "schema": {"f
ormat": "date", "type": "string"}}, "date_to": {"nullable": true, "required": false, "schema": {"format":
"date", "type": "string"}}, "entity_id": {"nullable": true, "required": false, "schema": {"type": "strin
g"}}, "entity_type": {"nullable": true, "required": false, "schema": {"type": "string"}}, "event_type": {
"nullable": true, "required": false, "schema": {"type": "string"}}}, "responses": {"200": {"schema": {"i
tems": {"additional_properties": true, "fields": {"actor": {"nullable": false, "required": true, "schema"
: {"type": "string"}}, "actor_id": {"nullable": true, "required": true, "schema": {"type": "string"}}, "a
ctor_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "after": {"nullable": tr
ue, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "at": {
"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "audit_id": {"
nullable": false, "required": true, "schema": {"type": "integer"}}, "before": {"nullable": true, "require
d": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "detail": {"nullabl
e": true, "required": true, "schema": {"type": "string"}}, "entity": {"nullable": false, "required": true
, "schema": {"type": "string"}}, "entity_id": {"nullable": false, "required": true, "schema": {"type": "s
tring"}}, "entity_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "event": {"
nullable": false, "required": true, "schema": {"type": "string"}}, "event_type": {"nullable": false, "req
uired": true, "schema": {"type": "string"}}, "occurred_at": {"nullable": false, "required": true, "schema
": {"format": "date-time", "type": "string"}}, "type": "object"}, "type": "array"}, "shape": "array"}, "
422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": fal
se, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required":
true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "typ
e": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"null
able": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "t
ype": "object"}, "shape": "object"}}
```

11 GET /dataset/bounds

RULES []

deferred inventory

12 GET /dashboard/summary

RULES []

```
{"request": {"body": null, "header": {}, "path": {}, "query": {"area": {"nullable": true, "required": fal
se, "schema": {"enum": ["ALL", "Etch", "Photo"], "type": "string"}}, "chamber_id": {"nullable": true, "re
quired": false, "schema": {"type": "string"}}, "date": {"nullable": true, "required": false, "schema": {"f
ormat": "date", "type": "string"}}, "equipment_id": {"nullable": true, "required": false, "schema": {"ty
pe": "string"}}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"alarm_cou
nt": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "area": {"null
able": true, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "daily_trend": {
"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"dat
e": {"nullable": false, "required": true, "schema": {"format": "date", "type": "string"}}, "has_r03_conse
c": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "ooc_count": {"nullable": false
, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "oos_count": {"nullable": false, "req
uired": true, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object"}, "type": "array"}}, "dat
e_range": {"nullable": false, "required": true, "schema": {"items": {"format": "date", "type": "string"},
"type": "array"}}, "equipment_counts": {"nullable": false, "required": true, "schema": {"items": {"addit
ional_properties": false, "fields": {"alarm_count": {"nullable": false, "required": true, "schema": {"min
imum": 0.0, "type": "integer"}}, "area_id": {"nullable": false, "required": true, "schema": {"enum": ["Et
ch", "Photo"], "type": "string"}}, "chambers": {"nullable": false, "required": true, "schema": {"items":
{"additional_properties": false, "fields": {"alarm_count": {"nullable": false, "required": true, "schema"
: {"minimum": 0.0, "type": "integer"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"m
in_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "equipment_id": {"nullable": fa
lse, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array
"}}, "hierarchy": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": fal
se, "fields": {"area_id": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "ty
pe": "string"}}, "chambers": {"nullable": false, "required": true, "schema": {"items": {"min_length": 1,
"type": "string"}, "type": "array"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"m
```

```

"length": 1, "type": "string"}, "type": "object"}, "type": "array"}}, "ooc_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "oos_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "pending_approvals": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"action_code": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "incident": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "request_id": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "severity": {"nullable": false, "required": true, "schema": {"enum": ["HIGH", "LOW", "MEDIUM"], "type": "string"}}, "type": "object"}, "type": "array"}}, "recent_alarms": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"action_code": {"nullable": true, "required": false, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "agent_run_status": {"nullable": true, "required": false, "schema": {"enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}}, "alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_status": {"nullable": true, "required": false, "schema": {"enum": ["APPROVED", "AUTO", "EXPIRED", "PENDING", "REJECTED"], "type": "string"}}, "area": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "detail": {"nullable": false, "required": true, "schema": {"type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "hit_cnt": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}}, "incident": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "judgement": {"nullable": false, "required": true, "schema": {"enum": ["OOC", "OOS"], "type": "string"}}, "latest_agent_run_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "lot_hist_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "occurred_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "recipe_step_name": {"nullable": true, "required": false, "schema": {"type": "string"}}, "recipe_step_no": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "rule_id": {"nullable": false, "required": true, "schema": {"enum": ["R01_OOS", "R02_OOC", "R03_CONSEC"], "type": "string"}}, "sensor_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "wafer_no": {"nullable": true, "required": false, "schema": {"minimum": 1.0, "type": "integer"}}, "type": "object"}, "type": "array"}}, "reference_date": {"nullable": false, "required": true, "schema": {"format": "date", "type": "string"}}, "sensor_catalog": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "type": "string"}, "top_sensors": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"alarm_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"items": {"min_length": 1, "type": "string"}, "type": "array"}}, "sensor_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "object"}, "422": {"schema": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}], "type": "array"}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "object"}}}}
13 GET /alarms/{source}/{alarm_id}
RULES []
{"request": {"body": null, "header": {}, "path": {"alarm_id": {"nullable": false, "required": true, "schema": {"type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "query": {}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"action_code": {"nullable": true, "required": false, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "agent_run_status": {"nullable": true, "required": false, "schema": {"enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}}, "alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_status": {"nullable": true, "required": false, "schema": {"enum": ["APPROVED", "AUTO", "EXPIRED", "PENDING", "REJECTED"], "type": "string"}}, "area": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "detail": {"nullable": false, "required": true, "schema": {"type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "hit_cnt": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}}, "incident": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "judgement": {"nullable": false, "required": true, "schema": {"enum": ["OOC", "OOS"], "type": "string"}}, "latest agent

```

```
ent_run_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "lot_hist_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "occurred_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "recipe_step_name": {"nullable": true, "required": false, "schema": {"type": "string"}}, "recipe_step_no": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "rule_id": {"nullable": false, "required": true, "schema": {"enum": ["R01_OOS", "R02_OOC", "R03_CONSEC"], "type": "string"}}, "sensor_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "wafer_no": {"nullable": true, "required": false, "schema": {"minimum": 1.0, "type": "integer"}}, "type": "object", "shape": "object", "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object", "shape": "object"}}
```

14 GET /alarms/paged

RULES []

```
{ "request": { "body": null, "header": {}, "path": {}, "query": { "area": {"nullable": true, "required": false, "schema": {"enum": ["ALL", "Etch", "Photo"], "type": "string"}}, "chamber_id": {"nullable": true, "required": false, "schema": {"type": "string"}}, "date": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "equipment_id": {"nullable": true, "required": false, "schema": {"type": "string"}}, "judgement": {"nullable": true, "required": false, "schema": {"enum": ["OOC", "OOS"], "type": "string"}}, "page": {"nullable": false, "required": false, "schema": {"default": 1, "minimum": 1, "type": "integer"}}, "sensor_id": {"nullable": true, "required": false, "schema": {"type": "string"}}, "size": {"nullable": false, "required": false, "schema": {"default": 20, "maximum": 100, "minimum": 1, "type": "integer"}}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"items": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"action_code": {"nullable": true, "required": false, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "agent_run_status": {"nullable": true, "required": false, "schema": {"enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}}, "alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_status": {"nullable": true, "required": false, "schema": {"enum": ["APPROVED", "AUTO", "EXPIRED", "PENDING", "REJECTED"], "type": "string"}}, "area": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "detail": {"nullable": false, "required": true, "schema": {"type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "hit_cnt": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}}, "incident": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_hist_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "occurred_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "recipe_step_name": {"nullable": true, "required": false, "schema": {"type": "string"}}, "recipe_step_no": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "rule_id": {"nullable": false, "required": true, "schema": {"enum": ["R01_OOS", "R02_OOC", "R03_CONSEC"], "type": "string"}}, "sensor_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "wafer_no": {"nullable": true, "required": false, "schema": {"minimum": 1.0, "type": "integer"}}, "type": "object", "type": "array"}, "page": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "size": {"nullable": false, "required": true, "schema": {"maximum": 100.0, "minimum": 1.0, "type": "integer"}}, "total": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object", "shape": "object", "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object", "shape": "object"}}
```

15 GET /traces/catalog

RULES []

```
{ "request": { "body": null, "header": {}, "path": {}, "query": {}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"areas": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"area_id": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "type": "object", "type": "array"}, "equipments": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"area_id": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chambers": {"nullable": false, "required": true, "schema": {"items": {"min_length": 1, "type": "string", "type": "array"}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object", "type": "array"}, "lots": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"lot_id": {"nullable": false, "required": true, "s
```

```
chema": {"min_length": 1, "type": "string"}}, "wafer_nos": {"nullable": false, "required": true, "schema": {"items": {"type": "integer"}, "type": "array"}}, "type": "object", "type": "array"}, "recipes": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"area_id": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "recipe_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "sensors": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"ctrl_lower": {"nullable": true, "required": false, "schema": {"type": "number"}}, "ctrl_upper": {"nullable": false, "required": true, "schema": {"type": "number"}}, "sensor_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "sensor_name": {"nullable": false, "required": true, "schema": {"type": "string"}}, "spec_lower": {"nullable": true, "required": false, "schema": {"type": "number"}}, "spec_upper": {"nullable": false, "required": true, "schema": {"type": "number"}}, "target": {"nullable": true, "required": false, "schema": {"type": "number"}}, "unit": {"nullable": true, "required": false, "schema": {"type": "string"}}, "upper_only": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "type": "object", "type": "array"}}, "type": "object"}, "shape": "object"}}
```

16 POST /traces/search

RULES []

```
{ "request": {"body": {"additional_properties": false, "fields": {"area": {"nullable": true, "required": false, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chamber_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "from": {"nullable": true, "required": false, "schema": {"format": "date-time", "type": "string"}}, "lot_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "recipe_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "sensor_ids": {"nullable": true, "required": false, "schema": {"items": {"min_length": 1, "type": "string"}, "type": "array"}}, "to": {"nullable": true, "required": false, "schema": {"format": "date-time", "type": "string"}}, "wafer_nos": {"nullable": true, "required": false, "schema": {"items": {"type": "integer"}, "type": "array"}}, "type": "object", "header": {}, "path": {}, "query": {}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"limits": {"nullable": false, "required": true, "schema": {"additional_properties": {"additional_properties": false, "fields": {"ctrl_lower": {"nullable": true, "required": false, "schema": {"type": "number"}}, "ctrl_upper": {"nullable": false, "required": true, "schema": {"type": "number"}}, "sensor_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "sensor_name": {"nullable": false, "required": true, "schema": {"type": "string"}}, "spec_lower": {"nullable": true, "required": false, "schema": {"type": "number"}}, "spec_upper": {"nullable": false, "required": true, "schema": {"type": "number"}}, "target": {"nullable": true, "required": false, "schema": {"type": "number"}}, "unit": {"nullable": true, "required": false, "schema": {"type": "string"}}, "upper_only": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "type": "object", "fields": {}, "type": "object"}, "measured_step_stats": {"nullable": false, "required": true, "schema": {"additional_properties": {"additional_properties": true, "fields": {}, "type": "object"}, "fields": {}, "type": "object"}, "total": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "wafers": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_hist_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "missing_steps": {"nullable": false, "required": false, "schema": {"items": {"type": "string"}, "type": "array"}}, "occurred_at": {"nullable": true, "required": false, "schema": {"format": "date-time", "type": "string"}}, "points": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"measured_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "recipe_step_name": {"nullable": true, "required": false, "schema": {"type": "string"}}, "recipe_step_no": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "seq_no": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "value": {"nullable": false, "required": true, "schema": {"type": "number"}}, "type": "object", "type": "array"}}, "sensor_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "wafer_no": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "type": "object", "type": "array"}}, "type": "object", "shape": "object"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "array"}}, "type": "object", "shape": "object"}}
```

17 GET /relations/equipment/{equipment_id}

RULES []

deferred inventory

18 GET /documents/{document_id}

RULES []

```
{ "request": {"body": null, "header": {}, "path": {"document_id": {"nullable": false, "required": true, "schema": {"type": "string"}}, "query": {}}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"chunks": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chunk_seq": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "content": {"nullable": false, "required": true, "schema": {"type": "string"}}, "section_title": {"nullable": true, "required": false, "schema": {"type": "string"}}, "type": "object", "type": "array"}}, "doc_type": {"nullable": true, "required": false, "schema": {"enum": ["MANUAL", "SPEC", "TROUBLESHOOT
```



```
], "type": "string"}}, {"document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"model_code": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, {"source_path": {"nullable": true, "required": false, "schema": {"type": "string"}}, {"title": {"nullable": false, "required": true, "schema": {"type": "string"}}, {"version": {"nullable": true, "required": false, "schema": {"type": "string"}}, {"type": "object"}, {"shape": "object"}, {"422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}, {"msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, {"type": {"nullable": false, "required": true, "schema": {"type": "string"}}, {"type": "object"}, {"type": "array"}}, {"type": "object"}, {"shape": "object"}}}}
```

19 POST /agent/runs

RULES []

```
{"request": {"body": {"additional_properties": false, "fields": {"alarm": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, {"type": "object"}}, {"type": "object"}, {"header": {"path": {}, "query": {}, "responses": {"202": {"schema": {"additional_properties": false, "fields": {"agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"alarm": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, {"type": "object"}}, {"status": {"nullable": false, "required": true, "schema": {"type": "string"}}, {"type": "object"}, {"shape": "object"}, {"422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}, {"msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, {"type": {"nullable": false, "required": true, "schema": {"type": "string"}}, {"type": "object"}, {"type": "array"}}, {"type": "object"}, {"shape": "object"}}}}
```

20 GET /agent/runs/{run_id}

RULES []

```
{"request": {"body": null, "header": {}, "path": {"run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"query": {}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"action": {"nullable": true, "required": true, "schema": {"additional_properties": false, "fields": {"action_code": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, {"action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"approval_status": {"nullable": true, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, {"deliveries": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES"], "type": "string"}}, {"completed_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, {"started_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, {"status": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, {"type": "object"}, {"type": "array"}}, {"reason": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"type": "object"}}, {"action_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"alarm_source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, {"approval": {"nullable": true, "required": true, "schema": {"additional_properties": false, "fields": {"action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"approval_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"decided_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, {"decided_by": {"nullable": true, "required": true, "schema": {"type": "string"}}, {"decision_comment": {"nullable": true, "required": true, "schema": {"type": "string"}}, {"status": {"nullable": false, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, {"type": "object"}}, {"approval_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"autonomy_level": {"nullable": false, "required": false, "schema": {"default": 2, "maximum": 3.0, "minimum": 1.0, "type": "integer"}}, {"chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"confidence": {"nullable": true, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, {"created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, {"deliveries": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES"], "type": "string"}}, {"status": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, {"type": "object"}, {"type": "array"}}, {"diagnosis": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"alternative_hypotheses": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"lower_rank_reason": {"nullable": false, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, {"summary": {"nullable": false, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, {"type": "object"}, {"type": "array"}}, {"cause_summary": {"nullable": true, "required": false, "schema": {"max_length": 2000, "type": "string"}}, {"confidence": {"n
```

ullable": true, "required": false, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "diagnostic_coverage": {"nullable": true, "required": false, "schema": {"type": "string"}}, "evidence_synthesis": {"nullable": true, "required": false, "schema": {"max_length": 2000, "type": "string"}}, "limitations": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "observations": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "origin_assessment": {"nullable": true, "required": false, "schema": {"additional_properties": false, "fields": {"basis": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"id": {"nullable": false, "required": true, "schema": {"max_length": 160, "min_length": 1, "type": "string"}}, "namespace": {"nullable": false, "required": true, "schema": {"enum": ["ALARM", "CHUNK", "LOT_HIST", "PARAMETER", "RELATION"]}, "type": "string"}}, "type": "object"}, "type": "array"}}, "compared": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"downstream": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"]}, "type": "string"}}, "history": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"]}, "type": "string"}}, "metrology": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"]}, "type": "string"}}, "sibling": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"]}, "type": "string"}}, "upstream": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"]}, "type": "string"}}, "scope": {"nullable": false, "required": true, "schema": {"enum": ["CURRENT_CHAMBER", "DOWNSTREAM", "EQUIPMENT_COMMON", "UNDETERMINED", "UPSTREAM"], "type": "string"}}, "type": "object"}, "parameter_findings": {"nullable": false, "required": true, "schema": {"default": [], "items": {"additional_properties": false, "fields": {"direction": {"nullable": false, "required": true, "schema": {"enum": ["ABOVE", "BELOW", "BOTH"], "type": "string"}}, "excursion_ratio": {"nullable": false, "required": true, "schema": {"type": "number"}}, "lot_hist_ids": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "parameter_id": {"nullable": false, "required": true, "schema": {"max_length": 20, "min_length": 1, "type": "string"}}, "step_no": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "wafer_scope": {"nullable": false, "required": true, "schema": {"enum": ["ALL", "PARTIAL", "SINGLE"], "type": "string"}}, "type": "object"}, "type": "array"}}, "predicted_fault_code": {"nullable": true, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "reason_code": {"nullable": true, "required": false, "schema": {"pattern": "^[A-Z0-9_]+\$", "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["AVAILABLE", "EMPTY"], "type": "string"}}, "verification_steps": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "type": "object"}, "evidence_assessment": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"available_sources": {"nullable": false, "required": true, "schema": {"items": {"enum": ["FDC", "GRAPH", "POSTGRES_ROUTE", "RAG"], "type": "string"}, "type": "array"}}, "conflicting_source_ids": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "missing_sources": {"nullable": false, "required": true, "schema": {"items": {"enum": ["FDC", "GRAPH", "POSTGRES_ROUTE", "RAG"], "type": "string"}, "type": "array"}}, "reason_codes": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["CONFLICT", "EMPTY", "PARTIAL", "SUFFICIENT"], "type": "string"}}, "type": "object"}, "evidence_items": {"nullable": false, "required": true, "schema": {"items": {"discriminator": "type", "type": "discriminated_union", "variants": {"ALARM": {"additional_properties": false, "fields": {"alarm": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "type": "object"}}, "excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "DOCUMENT": {"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "section": {"nullable": true, "required": true, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "METROLOGY": {"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}}, "type": "array"}}, "fault_code": {"nullable": true, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "fault_color": {"nullable": false, "required": true, "schema": {"type": "null"}}, "fault

```
{
  "name": "nullability: false, \"required\": true, \"schema\": {\"type\": \"null\"}}, \"impact_scope\": {\"nullable: f\nfalse, \"required\": true, \"schema\": {\"additional_properties\": false, \"fields\": {\"check_required\": {\"nullabl\ne\": false, \"required\": true, \"schema\": {\"items\": {\"additional_properties\": false, \"fields\": {\"kind\": {\"nu\nllable\": false, \"required\": true, \"schema\": {\"enum\": [\"CHAMBER\", \"LOT\", \"PARAMETER\", \"PROCESS_STEP\", \"SIB\nLING_CHAMBER\", \"WAFER\"]}, \"type\": \"string\"}}}, \"relation\": {\"nullable\": true, \"required\": false, \"schema\":\n{\"type\": \"string\"}}, \"source_id\": {\"nullable\": false, \"required\": true, \"schema\": {\"min_length\": 1, \"type\n\": \"string\"}}, \"type\": \"object\", \"type\": \"array\"}}, \"direct\": {\"nullable\": false, \"required\": true, \"sc\nhema\": {\"items\": {\"additional_properties\": false, \"fields\": {\"kind\": {\"nullable\": false, \"required\": true\n\", \"schema\": {\"enum\": [\"CHAMBER\", \"LOT\", \"PARAMETER\", \"PROCESS_STEP\", \"SIBLING_CHAMBER\", \"WAFER\"]}, \"type\":\n\"string\"}}, \"relation\": {\"nullable\": true, \"required\": false, \"schema\": {\"type\": \"string\"}}, \"source_id\n\": {\"nullable\": false, \"required\": true, \"schema\": {\"min_length\": 1, \"type\": \"string\"}}, \"type\": \"object\n\", \"type\": \"array\"}}, \"graph_conflict\": {\"nullable\": false, \"required\": false, \"schema\": {\"default\": fals\ne, \"type\": \"boolean\"}}, \"reason_code\": {\"nullable\": true, \"required\": false, \"schema\": {\"pattern\": \"^[A-Z\n0-9_]+$\", \"type\": \"string\"}}, \"status\": {\"nullable\": false, \"required\": true, \"schema\": {\"enum\": [\"AVAILA\nBLE\", \"EMPTY\"], \"type\": \"string\"}}, \"summary\": {\"nullable\": true, \"required\": false, \"schema\": {\"max_leng\nth\": 2000, \"type\": \"string\"}}, \"type\": \"object\"}}, \"latency_ms\": {\"nullable\": false, \"required\": true, \"\nschema\": {\"minimum\": 0.0, \"type\": \"integer\"}}, \"llm_model\": {\"nullable\": false, \"required\": true, \"schema\n\": {\"min_length\": 1, \"type\": \"string\"}}, \"post_action_observation\": {\"nullable\": false, \"required\": true,\n\"schema\": {\"additional_properties\": false, \"fields\": {\"message\": {\"nullable\": false, \"required\": false,\n\"schema\": {\"default\": \"\\ucd5c\\uc885 \\uc815\\uc801 \\ub370\\uc774\\ud130\\uc14b\\uc5d0\\ub294 \\uc870\\uce58 \\uc774\n\\ud6c4 \\uacf5\\uc815 \\uad00\\uce21\\uac12\\uc774 \\uc5c6\\uc5b4 \\ud6a8\\uacfc\\ub97c \\ud3c9\\uac00\\ud560 \\uc218 \\u\n5c6\\uc74c\", \"type\": \"string\"}}, \"status\": {\"nullable\": false, \"required\": false, \"schema\": {\"default\": \"\nNOT_AVAILABLE_STATIC_DATASET\"}, \"type\": \"string\"}}, \"type\": \"object\"}}, \"predicted_fault_code\": {\"nullabl\ne\": true, \"required\": true, \"schema\": {\"enum\": [\"FOC\", \"MFD\", \"OTH\", \"RFM\", \"TMD\"], \"type\": \"string\"}}, \"\nprediction\": {\"nullable\": true, \"required\": true, \"schema\": {\"additional_properties\": false, \"fields\": {\"ca\nuse_summary\": {\"nullable\": false, \"required\": true, \"schema\": {\"max_length\": 2000, \"min_length\": 1, \"ty\npe\": \"string\"}}, \"confidence\": {\"nullable\": false, \"required\": true, \"schema\": {\"maximum\": 1.0, \"minimum\"\n: 0.0, \"type\": \"number\"}}, \"generated_at\": {\"nullable\": false, \"required\": true, \"schema\": {\"format\": \"da\nte-time\", \"type\": \"string\"}}, \"input_tokens\": {\"nullable\": false, \"required\": true, \"schema\": {\"minimum\":\n0.0, \"type\": \"integer\"}}, \"llm_model\": {\"nullable\": false, \"required\": true, \"schema\": {\"min_length\": 1,\n\"type\": \"string\"}}, \"output_tokens\": {\"nullable\": false, \"required\": true, \"schema\": {\"minimum\": 0.0, \"t\nype\": \"integer\"}}, \"predicted_fault_code\": {\"nullable\": false, \"required\": true, \"schema\": {\"enum\": [\"FOC\n\", \"MFD\", \"OTH\", \"RFM\", \"TMD\"], \"type\": \"string\"}}, \"prompt_version\": {\"nullable\": false, \"required\": tru\ne, \"schema\": {\"min_length\": 1, \"type\": \"string\"}}, \"supporting_alarms\": {\"nullable\": false, \"required\": t\nue, \"schema\": {\"items\": {\"additional_properties\": false, \"fields\": {\"alarm_id\": {\"nullable\": false, \"req\nquired\": true, \"schema\": {\"min_length\": 1, \"type\": \"string\"}}, \"source\": {\"nullable\": false, \"required\": t\nue, \"schema\": {\"enum\": [\"R03\", \"SUMMARY\", \"TRACE\"], \"type\": \"string\"}}, \"type\": \"object\", \"type\": \"arr\nay\"}}, \"supporting_chunk_ids\": {\"nullable\": false, \"required\": true, \"schema\": {\"items\": {\"min_length\": 1\n\", \"type\": \"string\"}, \"type\": \"array\"}}, \"supporting_relation_ids\": {\"nullable\": false, \"required\": true,\n\"schema\": {\"items\": {\"min_length\": 1, \"type\": \"string\"}, \"type\": \"array\"}}, \"uncertainty\": {\"nullable\": f\nalse, \"required\": true, \"schema\": {\"max_length\": 1000, \"type\": \"string\"}}, \"type\": \"object\"}}, \"react_tr\nace\": {\"nullable\": false, \"required\": false, \"schema\": {\"items\": {\"additional_properties\": false, \"fields\n\": {\"argument_summary\": {\"nullable\": true, \"required\": false, \"schema\": {\"pattern\": \"^(\\ud6c4\\ubcf4 wafer\n[1-9][0-9]*\\\\((\\ud604\\uc7ac\\uc0c1\\ub958|\\ud558\\ub958)\\\\)\\\\)\\uc774\\ub825 [1-9][0-9]*\\\\((\\ud604\\uc7ac\\ud61\n5\\uc81c)\\\\)\\\\)\\uacc4\\uce21 [1-9][0-9]*\\\\((\\ud604\\uc7ac\\uc0c1\\ub958|\\ud558\\ub958)\\\\)\\\\)\\ubb38\\uc11c \\uac80\\uc\n0c9\\uc124\\ube44 \\ucee8\\ud14d\\uc2a4\\ud2b8)$\", \"type\": \"string\"}}, \"degraded\": {\"nullable\": false, \"requir\ned\": false, \"schema\": {\"default\": false, \"type\": \"boolean\"}}, \"guard_code\": {\"nullable\": true, \"required\n\": false, \"schema\": {\"max_length\": 64, \"type\": \"string\"}}, \"observation_summary\": {\"nullable\": true, \"requ\nired\": false, \"schema\": {\"max_length\": 160, \"type\": \"string\"}}, \"phase\": {\"nullable\": false, \"required\":\ntrue, \"schema\": {\"enum\": [\"OBSERVED\", \"REJECTED\", \"SELECTED\", \"STOPPED\"], \"type\": \"string\"}}, \"rationale_\nsummary\": {\"nullable\": true, \"required\": false, \"schema\": {\"max_length\": 120, \"type\": \"string\"}}, \"react_\nprompt_version\": {\"nullable\": true, \"required\": false, \"schema\": {\"type\": \"string\"}}, \"selector_tokens\":\n{\"nullable\": false, \"required\": true, \"schema\": {\"additional_properties\": false, \"fields\": {\"input\": {\"nu\nllable\": false, \"required\": true, \"schema\": {\"minimum\": 0.0, \"type\": \"integer\"}}, \"output\": {\"nullable\n\": false, \"required\": true, \"schema\": {\"minimum\": 0.0, \"type\": \"integer\"}}, \"type\": \"object\"}}, \"seq\": {\"nu\nllable\": false, \"required\": true, \"schema\": {\"minimum\": 1.0, \"type\": \"integer\"}}, \"stop_reason\": {\"nullab\nle\": true, \"required\": false, \"schema\": {\"enum\": [\"BUDGET_EXHAUSTED\", \"GUARD_LIMIT\", \"LLM_DEPENDENCY\", \"L\nLM_STOP\", \"LLM_TIMEOUT\", \"REACT_STRUCTURE_INVALID\", \"STEP_CAP\", \"type\": \"string\"}}, \"tool\": {\"nullable\":\ntrue, \"required\": false, \"schema\": {\"enum\": [\"get_chamber_parameter_history\", \"get_equipment_context\", \"g\net_fdc_summary\", \"get_metrology_result\", \"search_documents\", \"stop\"], \"type\": \"string\"}}, \"type\": \"obje\ncq\"}, \"type\": \"array\"}}, \"recommended_action\": {\"nullable\": true, \"required\": true, \"schema\": {\"enum\": [\"E\nQP_HOLD\", \"MONITORING\", \"WARNING\"], \"type\": \"string\"}}, \"remaining_read_calls\": {\"nullable\": false, \"req\nquired\": false, \"schema\": {\"default\": 0, \"maximum\": 8.0, \"minimum\": 0.0, \"type\": \"integer\"}}, \"similar_inc\nidents\": {\"nullable\": false, \"required\": true, \"schema\": {\"additional_properties\": false, \"fields\": {\"ite\nms\": {\"nullable\": false, \"required\": true, \"schema\": {\"items\": {\"additional_properties\": false, \"fields\":\n{\"agent_run_id\": {\"nullable\": false, \"required\": true, \"schema\": {\"min_length\": 1, \"type\": \"string\"}}, \"\nchamber_id\": {\"nullable\": false, \"required\": true, \"schema\": {\"min_length\": 1, \"type\": \"string\"}}, \"lot_i\nd\": {\"nullable\": false, \"required\": true, \"schema\": {\"min_length\": 1, \"type\": \"string\"}}, \"parameter_jacc\nard\": {\"nullable\": false, \"required\": true, \"schema\": {\"maximum\": 1.0, \"minimum\": 0.0, \"type\": \"number\"}}\n\", \"predicted_fault_code\": {\"nullable\": false, \"required\": true, \"schema\": {\"enum\": [\"FOC\", \"MFD\", \"OTH\n\", \"RFM\", \"TMD\"], \"type\": \"string\"}}, \"recommended_action\": {\"nullable\": false, \"required\": true, \"schema\n\": {\"enum\": [\"EQP_HOLD\", \"MONITORING\", \"WARNING\"], \"type\": \"string\"}}, \"score\": {\"nullable\": false, \"require\nd\": true, \"schema\": {\"maximum\": 100.0, \"minimum\": 0.0, \"type\": \"integer\"}}, \"type\": \"object\", \"type\": \"\narray\"}}, \"label\": {\"nullable\": false, \"required\": false, \"schema\": {\"default\": \"uace0\\uc815 \\uc2dc\\uc5f
```

```

0\b370\udc774\ud130\ub0b4\ube44a\ud505\ua300\uacfc", "type": "string"}, {"reason_code": {"nullable": true, "required": false, "schema": {"pattern": "^[A-Z0-9_]+$", "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["AVAILABLE", "EMPTY"], "type": "string"}}, "type": "object"}}, {"status": {"nullable": false, "required": true, "schema": {"enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}}, "tools": {"nullable": false, "required": true, "schema": {"item_s": {"additional_properties": false, "fields": {"n": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result_summary": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "s": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "tool_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "trace_state": {"nullable": false, "required": false, "schema": {"default": "NOT_APPLICABLE", "enum": ["AVAILABLE", "NOT_APPLICABLE", "PENDING", "UNAVAILABLE"], "type": "string"}}, "type": "object"}, "shape": "object"}, "404": {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"enum": ["APPROVAL_ALREADY_DECIDED", "DEPENDENCY_NOT_READY", "IDEMPOTENCY_CONFLICT", "INCIDENT_ALREADY_PROCESSED", "INCIDENT_ALREADY_RUNNING", "INTERNAL_ERROR", "LEGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_NOT_FOUND", "UNAUTHORIZED", "VALIDATION_ERROR"], "type": "string"}}, "details": {"nullable": false, "required": false, "schema": {"additional_properties": true, "fields": {"type": "object"}}, "message": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "shape": "object"}, "422": {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"enum": ["APPROVAL_ALREADY_DECIDED", "DEPENDENCY_NOT_READY", "IDEMPOTENCY_CONFLICT", "INCIDENT_ALREADY_PROCESSED", "INCIDENT_ALREADY_RUNNING", "INTERNAL_ERROR", "LEGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_NOT_FOUND", "UNAUTHORIZED", "VALIDATION_ERROR"], "type": "string"}}, "details": {"nullable": false, "required": false, "schema": {"additional_properties": true, "fields": {"type": "object"}}, "message": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "shape": "object"}}, {"503": {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"enum": ["APPROVAL_ALREADY_DECIDED", "DEPENDENCY_NOT_READY", "IDEMPOTENCY_CONFLICT", "INCIDENT_ALREADY_PROCESSED", "INCIDENT_ALREADY_RUNNING", "INTERNAL_ERROR", "LEGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_NOT_FOUND", "UNAUTHORIZED", "VALIDATION_ERROR"], "type": "string"}}, "details": {"nullable": false, "required": false, "schema": {"additional_properties": true, "fields": {"type": "object"}}, "message": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "shape": "object"}}, {"21 GET /agent/evaluations RULES [{"fault_5class": {"nullable": true, "required": true, "schema": {"additional_properties": false, "fields": {"classification": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"accuracy": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"denominator": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "numerator": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "rate": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "type": "object"}, "by_class": {"nullable": false, "required": true, "schema": {"additional_properties": {"additional_properties": false, "fields": {"f1": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "false_negative": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "false_positive": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "precision": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "recall": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "support": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "true_positive": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object"}, "fields": {"type": "object"}, "macro_f1_5class": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "observed_class_macro_f1": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "population_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "unclassified_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object"}, "evdence_valid_run": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"denominator": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "numerator": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "rate": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "type": "object"}, "exclusions": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "meaning": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "reason": {"nullable": false, "required": true, "schema": {"enum": ["AMBIGUOUS_LABEL", "NO_INJECTED_FAULT"], "type": "string"}}, "type": "object"}, "type": "array"}}, "hard_gate_passed": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "hard_gate_reasons": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "label_source": {"nullable": false, "required": true, "schema": {"type": "string"}}, "metrology_observed_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "metrology_total_lot_hist_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "production_ground_truth_available": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "production_performance_disclaimer": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "public

```

```
_fault_ground_truth_available": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "rule_action_agreement": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"denominator": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "numerator": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "rate": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "type": "object"}}, "structured_prediction": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"denominator": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "numerator": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "rate": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "type": "number"}}, "usage_scope": {"nullable": false, "required": true, "schema": {"type": "string"}}, "versions": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"dataset_epoch": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "model_version": {"nullable": true, "required": true, "schema": {"type": "string"}}, "policy_version": {"nullable": true, "required": true, "schema": {"type": "string"}}, "prompt_version": {"nullable": true, "required": true, "schema": {"type": "string"}}, "type": "object"}}, "type": "object"}}, "fault_5class_empty_reason": {"nullable": true, "required": true, "schema": {"enum": ["CONTRACT_VIOLATION", "NOT_CONFIGURED", "NOT_FOUND", "type": "string"]}}, "golden_flow": {"nullable": true, "required": true, "schema": {"dataset_epoch": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "phases": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"metrics": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "phase": {"nullable": false, "required": true, "schema": {"enum": ["BATCH_BASELINE", "DECISIONS", "MANUAL_RETRY", "PRE_FLIGHT", "PRE_APPROVAL", "SECOND_BATCH", "UNKNOWN", "type": "string"]}, "reasons": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["EVIDENCE_INCOMPLETE", "FAIL", "PASS", "type": "string"]}, "type": "object"}, "type": "array"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["EVIDENCE_INCOMPLETE", "FAIL", "PASS", "type": "string"]}, "type": "object"}}, "golden_flow_empty_reason": {"nullable": true, "required": true, "schema": {"enum": ["CONTRACT_VIOLATION", "NOT_CONFIGURED", "NOT_FOUND", "type": "string"]}, "type": "object"}, "shape": "object"}}, 22 POST /agent/runs/{run_id}/retry
RULES []
deferred inventory
23 GET /agent/runs/paged
RULES []
deferred inventory
24 GET /approvals/paged
RULES []
deferred inventory
25 GET /actions
RULES []
{"request": {"body": null, "header": {}, "path": {}, "query": {"action_code": {"nullable": true, "required": false, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING", "type": "string"]}}, "responses": {"200": {"schema": {"items": {"additional_properties": false, "fields": {"action_code": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING", "type": "string"]}, "action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_status": {"nullable": true, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED", "type": "string"]}, "chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "created_by_agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "deliveries": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES", "type": "string"]}, "status": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELLED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING", "type": "string"]}, "type": "object"}, "type": "array"}}, "equipment": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "reason": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}, "shape": "array"}, "422": {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"enum": ["APPROVAL_ALREADY_DECIDED", "DEPENDENCY_NOT_READY", "IDEMPOTENCY_CONFLICT", "INCIDENT_ALREADY_PROCESSED", "INCIDENT_ALREADY_RUNNING", "INTERNAL_ERROR", "LEGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_NOT_FOUND", "UNAUTHORIZED", "VALIDATION_ERROR", "type": "string"]}, "details": {"nullable": false, "required": false, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "message": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "shape": "object"}}, 503: {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"enum": ["APPROVAL_ALREADY_DECIDED", "DEPENDENCY_NOT_READY", "IDEMPOTENCY_CONFLICT", "INCIDENT_ALREADY_PROCESSED", "INCIDENT_ALREADY_RUNNING", "INTERNAL_ERROR", "LEGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_NOT_FOUND", "UNAUTHORIZED", "VALIDATION_ERROR", "type": "string"]}, "details": {"nullable": false, "required": false, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "message": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "shape": "object"}},
```

```
26 GET /actions/{action_id}
RULES []
{"request": {"body": null, "header": {}, "path": {"action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "query": {}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"action_code": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}, "action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "approval_status": {"nullable": true, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}, "chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}, "created_by_agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "deliveries": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES"], "type": "string"}, "completed_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}, "started_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}, "status": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, "type": "object"}, "type": "array"}}, "equipment": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}, "equipment_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}, "lot": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "reason": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "shape": "object", "404": {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"enum": ["APPROVAL_ALREADY_DECIDED", "DEPENDENCY_NOT_READY", "IDEMPOTENCY_CONFLICT", "INCIDENT_ALREADY_PROCESSED", "INCIDENT_ALREADY_RUNNING", "INTERNAL_ERROR", "LEGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_NOT_FOUND", "UNAUTHORIZED", "VALIDATION_ERROR"], "type": "string"}}, "details": {"nullable": false, "required": false, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "message": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "shape": "object", "422": {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"enum": ["APPROVAL_ALREADY_DECIDED", "DEPENDENCY_NOT_READY", "IDEMPOTENCY_CONFLICT", "INCIDENT_ALREADY_PROCESSED", "INCIDENT_ALREADY_RUNNING", "INTERNAL_ERROR", "LEGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_NOT_FOUND", "UNAUTHORIZED", "VALIDATION_ERROR"], "type": "string"}, "details": {"nullable": false, "required": false, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "message": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "shape": "object", "503": {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"enum": ["APPROVAL_ALREADY_DECIDED", "DEPENDENCY_NOT_READY", "IDEMPOTENCY_CONFLICT", "INCIDENT_ALREADY_PROCESSED", "INCIDENT_ALREADY_RUNNING", "INTERNAL_ERROR", "LEGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_NOT_FOUND", "UNAUTHORIZED", "VALIDATION_ERROR"], "type": "string"}, "details": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "message": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "shape": "object"}}}}
27 POST /actions/{action_id}/deliveries/{channel}/retry
RULES []
deferred inventory
28 POST /internal/actions/{action_id}/delivery
RULES [{"FAILED requires error_code non-null and permits provider_message_id null", "SENT requires provider_message_id non-null and error_code null"}]
{"request": {"body": {"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES MOCK"], "type": "string"}, "completed_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}, "error_code": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}, "event_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "provider_message_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}, "request_hash": {"nullable": false, "required": true, "schema": {"max_length": 64, "min_length": 64, "pattern": "^[0-9a-f]{64}$", "type": "string"}, "status": {"nullable": false, "required": true, "schema": {"enum": ["FAILED", "SENT"], "type": "string"}}, "type": "object"}, "header": {"X-Delivery-Signature": {"nullable": true, "required": false, "schema": {"type": "string"}}, "X-Delivery-Timestamp": {"nullable": true, "required": false, "schema": {"type": "string"}}, "path": {"action_id": {"nullable": false, "required": true, "schema": {"type": "string"}}, "query": {}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"action_id": {"nullable": false, "required": true, "schema": {"max_length": 20, "min_length": 1, "type": "string"}, "channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES MOCK"], "type": "string"}, "completed_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}, "duplicate": {"nullable": false, "required": true, "schema": {"type": "boolean"}, "error_code": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}, "provider_message_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}, "request_hash": {"nullable": false, "required": true, "schema": {"max_length": 64, "min_length": 64, "pattern": "^[0-9a-f]{64}$", "type": "string"}, "status": {"nullable": false, "required": true, "schema": {"enum": ["FAILED", "SENT"], "type": "string"}}, "type": "object"}, "shape": "object", "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable":
```

```

false, "required": true, "schema": {"type": "string"}, "type": "nullable": false, "required": true, "sc
hema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "object"}
}}
29 POST /analytics/query
RULES []
{"request": {"body": {"additional_properties": false, "fields": {"question": {"nullable": false, "require
d": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "type": "object"}, "header
": {"path": {}, "query": {}}, "responses": {"200": {"schema": {"additional_properties": false, "fields
": {"columns": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "arr
ay"}}, "cross_check": {"nullable": true, "required": false, "schema": {"additional_properties": false, "f
ields": {"cypther": {"nullable": true, "required": false, "schema": {"type": "string"}, "status": {"nulla
ble": false, "required": true, "schema": {"enum": ["MATCH", "MISMATCH", "SKIPPED"], "type": "string"}}, "
summary": {"nullable": true, "required": false, "schema": {"type": "string"}}, "type": "object"}}, "erro
r_msg": {"nullable": true, "required": false, "schema": {"type": "string"}}, "generated_sql": {"nullable"
: true, "required": false, "schema": {"type": "string"}}, "group_by": {"nullable": false, "required": tru
e, "schema": {"items": {"type": "string"}, "type": "array"}}, "is_rejected": {"nullable": false, "require
d": true, "schema": {"type": "boolean"}}, "is_valid": {"nullable": false, "required": true, "schema": {"t
ype": "boolean"}}, "latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "typ
e": "integer"}}, "metric": {"nullable": true, "required": false, "schema": {"additional_properties": false,
"fields": {"column": {"nullable": true, "required": false, "schema": {"type": "string"}}, "p": {"nullabl
e": true, "required": false, "schema": {"maximum": 100.0, "minimum": 0.0, "type": "number"}}, "type": {"nu
llable": false, "required": true, "schema": {"enum": ["count", "max", "mean", "median", "min", "percentil
e", "ratio", "std", "sum"], "type": "string"}}, "type": "object"}}, "metric_result": {"nullable": false,
"required": false, "schema": {"type": "union", "variants": [{"type": "integer"}, {"type": "number"}, {"i
tems": {"additional_properties": false, "fields": {"group": {"nullable": false, "required": true, "schema
": {"additional_properties": true, "fields": {"type": "object"}, "value": {"nullable": false, "require
d": false, "schema": {"type": "union", "variants": [{"type": "integer"}, {"type": "number"}, {"type": "n
ull"}]}}, "type": "object"}, "type": "array"}, {"type": "null"}]}, "nl_query_log_id": {"nullable": true
, "required": false, "schema": {"minimum": 1.0, "type": "integer"}}, "question": {"nullable": false, "req
uired": true, "schema": {"type": "string"}}, "reject_reason": {"nullable": true, "required": false, "sche
ma": {"type": "string"}}, "row_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "
type": "integer"}}, "rows": {"nullable": false, "required": true, "schema": {"items": {"additional_proper
ties": true, "fields": {"type": "object"}, "type": "array"}}, "visualization": {"nullable": true, "req
uired": false, "schema": {"additional_properties": false, "fields": {"chart_type": {"nullable": false, "r
equired": true, "schema": {"enum": ["bar", "histogram", "line", "table"], "type": "string"}}, "x": {"null
able": true, "required": false, "schema": {"type": "string"}}, "y": {"nullable": true, "required": false,
"schema": {"type": "string"}}, "type": "object"}}, "type": "object"}, "shape": "object"}, "422": {"sch
ema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema
": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "sch
ema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"
}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": fals
e, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "obje
ct"}, "shape": "object"}}}
30 POST /analytics/graph-query
RULES []
{"request": {"body": {"additional_properties": false, "fields": {"question": {"nullable": false, "require
d": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "type": "object"}, "header
": {"path": {}, "query": {}}, "responses": {"200": {"schema": {"additional_properties": false, "fields
": {"columns": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "arr
ay"}}, "error_msg": {"nullable": true, "required": false, "schema": {"type": "string"}}, "generated_cyphe
r": {"nullable": true, "required": false, "schema": {"type": "string"}}, "is_rejected": {"nullable": fals
e, "required": true, "schema": {"type": "boolean"}}, "is_valid": {"nullable": false, "required": true, "s
chema": {"type": "boolean"}}, "latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0
.0, "type": "integer"}}, "question": {"nullable": false, "required": true, "schema": {"type": "string"}},
"reject_reason": {"nullable": true, "required": false, "schema": {"type": "string"}}, "row_count": {"nul
lable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "rows": {"nullable": fal
se, "required": true, "schema": {"items": {"additional_properties": true, "fields": {"type": "object"}
, "type": "array"}}, "type": "object"}, "shape": "object"}, "422": {"schema": {"additional_properties":
true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_proper
ties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union"
, "variants": [{"type": "string"}, {"type": "integer"}}, "type": "array"}}, "msg": {"nullable": false,
"required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {
"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "object"}}}
31 POST /analytics/validate
RULES []
{"request": {"body": {"additional_properties": false, "fields": {"sql": {"nullable": false, "required": t
rue, "schema": {"max_length": 20000, "min_length": 1, "type": "string"}}, "type": "object"}, "header": {
}, "path": {}, "query": {}}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {
"checks": {"nullable": true, "required": false, "schema": {"items": {"additional_properties": false, "fie
lds": {"key": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "labe
l": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "ok": {"nullabl
e": false, "required": true, "schema": {"type": "boolean"}}, "type": "object"}, "type": "array"}}, "norm
alized_sql": {"nullable": true, "required": false, "schema": {"type": "string"}}, "reason": {"nullable":
false, "required": true, "schema": {"type": "string"}}, "valid": {"nullable": false, "required": true, "s

```

```
chema": {"type": "boolean"}}, {"type": "object"}, {"shape": "object"}, "422": {"schema": {"additional_prop
erties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additi
al_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type
": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable":
false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "s
chema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object"}, {"shape": "object"
}}}
32 GET /analytics/history
RULES []
{"request": {"body": null, "header": {}, "path": {}, "query": {"date_from": {"nullable": true, "required"
: false, "schema": {"format": "date", "type": "string"}}, "date_to": {"nullable": true, "required": false
, "schema": {"format": "date", "type": "string"}}, "is_rejected": {"nullable": true, "required": false, "
schema": {"type": "boolean"}}, "is_valid": {"nullable": true, "required": false, "schema": {"type": "bool
ean"}}, "page": {"nullable": false, "required": false, "schema": {"default": 1, "minimum": 1, "type": "in
teger"}}, "size": {"nullable": false, "required": false, "schema": {"default": 20, "maximum": 100, "minim
um": 1, "type": "integer"}}, "responses": {"200": {"schema": {"additional_properties": false, "fields":
{"items": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fi
elds": {"asked_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "stri
ng"}}, "error_msg": {"nullable": true, "required": false, "schema": {"type": "string"}}, "generated_sql":
{"nullable": true, "required": false, "schema": {"type": "string"}}, "is_rejected": {"nullable": false,
"required": true, "schema": {"type": "boolean"}}, "is_valid": {"nullable": false, "required": true, "sche
ma": {"type": "boolean"}}, "latency_ms": {"nullable": true, "required": false, "schema": {"minimum": 0.0,
"type": "integer"}}, "nl_query_log_id": {"nullable": false, "required": true, "schema": {"minimum": 1.0,
"type": "integer"}}, "outcome": {"nullable": false, "required": true, "schema": {"enum": ["DB_ERROR", "P
OLICY_REJECTED", "SUCCESS", "VALIDATION_FAILED"], "type": "string"}}, "question": {"nullable": false, "re
quired": true, "schema": {"type": "string"}}, "reject_reason": {"nullable": true, "required": false, "sch
ema": {"type": "string"}}, "row_cnt": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "t
ype": "integer"}}, "type": "object"}, "type": "array"}}, "page": {"nullable": false, "required": true, "
schema": {"minimum": 1.0, "type": "integer"}, "size": {"nullable": false, "required": true, "schema": {"
maximum": 100.0, "minimum": 1.0, "type": "integer"}, "total": {"nullable": false, "required": true, "sch
ema": {"minimum": 0.0, "type": "integer"}}, "type": "object"}, {"shape": "object"}, "422": {"schema": {"a
dditional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"item
s": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"i
tems": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg"
: {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "requi
red": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object"}, {"sh
ape": "object"}}}}
33 GET /analytics/evaluations
RULES []
{"request": {"body": null, "header": {}, "path": {}, "query": {"latest": {"nullable": false, "required":
false, "schema": {"default": true, "type": "boolean"}}, "page": {"nullable": false, "required": false, "s
chema": {"default": 1, "minimum": 1, "type": "integer"}, "size": {"nullable": false, "required": false,
"schema": {"default": 20, "maximum": 100, "minimum": 1, "type": "integer"}}, "responses": {"200": {"sch
ema": {"additional_properties": false, "fields": {"items": {"nullable": false, "required": true, "schema"
: {"items": {"additional_properties": false, "fields": {"accuracy": {"nullable": false, "required": true,
"schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "correct": {"nullable": false, "required"
: true, "schema": {"minimum": 0.0, "type": "integer"}}, "defense_passed": {"nullable": false, "required":
true, "schema": {"minimum": 0.0, "type": "integer"}}, "defense_total": {"nullable": false, "required": t
rue, "schema": {"minimum": 0.0, "type": "integer"}}, "executed_at": {"nullable": false, "required": true,
"schema": {"format": "date-time", "type": "string"}}, "items": {"nullable": false, "required": true, "sc
hema": {"items": {"additional_properties": false, "fields": {"actual_result": {"nullable": true, "require
d": false, "schema": {"type": "any"}}, "actual_visualization": {"nullable": true, "required": false, "sch
ema": {"additional_properties": false, "fields": {"chart_type": {"nullable": false, "required": true, "sc
hema": {"enum": ["bar", "histogram", "line", "table"], "type": "string"}}, "x": {"nullable": true, "requi
red": false, "schema": {"type": "string"}}, "y": {"nullable": true, "required": false, "schema": {"type":
"string"}}, "type": "object"}}, "attempt_count": {"nullable": false, "required": true, "schema": {"mini
mum": 0.0, "type": "integer"}, "case_id": {"nullable": false, "required": true, "schema": {"min_length":
1, "type": "string"}}, "case_type": {"nullable": false, "required": true, "schema": {"enum": ["DEFENSE",
"GOLD"], "type": "string"}}, "expected_result": {"nullable": true, "required": false, "schema": {"type":
"any"}}, "expected_visualization": {"nullable": true, "required": false, "schema": {"additional_properti
es": false, "fields": {"chart_type": {"nullable": false, "required": true, "schema": {"enum": ["bar", "hi
stogram", "line", "table"], "type": "string"}}, "x": {"nullable": true, "required": false, "schema": {"ty
pe": "string"}}, "y": {"nullable": true, "required": false, "schema": {"type": "string"}}, "type": "obje
ct"}}, "generated_sql": {"nullable": true, "required": false, "schema": {"type": "string"}}, "latency_ms"
: {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}, "passed": {"nulla
ble": false, "required": true, "schema": {"type": "boolean"}}, "question": {"nullable": true, "required":
false, "schema": {"type": "string"}}, "reason": {"nullable": true, "required": false, "schema": {"type":
"string"}}, "type": "object"}, "type": "array"}}, "model": {"nullable": false, "required": true, "schem
a": {"min_length": 1, "type": "string"}, "prompt_version": {"nullable": false, "required": true, "schema
": {"min_length": 1, "type": "string"}, "provider": {"nullable": false, "required": true, "schema": {"mi
n_length": 1, "type": "string"}, "run_id": {"nullable": false, "required": true, "schema": {"min_length"
: 1, "type": "string"}, "temperature": {"nullable": false, "required": true, "schema": {"type": "number"
}}, "total": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "type
": "object"}, "type": "array"}}, "page": {"nullable": false, "required": true, "schema": {"minimum": 1.0,
```



```
"type": "integer"}}, "size": {"nullable": false, "required": true, "schema": {"maximum": 100.0, "minimum": 1.0, "type": "integer"}}, "total": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object", "shape": "object"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "array"}}, "type": "object"}, "shape": "object"}}
```

34 GET /audit-logs/paged

RULES []

```
{ "request": { "body": null, "header": {}, "path": {}, "query": { "actor_type": {"nullable": true, "required": false, "schema": {"type": "string"}}, "date_from": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "date_to": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "entity_id": {"nullable": true, "required": false, "schema": {"type": "string"}}, "entity_type": {"nullable": true, "required": false, "schema": {"type": "string"}}, "event_type": {"nullable": true, "required": false, "schema": {"type": "string"}}, "page": {"nullable": false, "required": false, "schema": {"default": 1, "minimum": 1, "type": "integer"}}, "size": {"nullable": false, "required": false, "schema": {"default": 20, "maximum": 100, "minimum": 1, "type": "integer"}}, "responses": {"200": {"schema": {"additional_properties": true, "fields": {"event_type_counts": {"nullable": false, "required": true, "schema": {"additional_properties": {"type": "integer"}, "fields": {}, "type": "object"}}, "event_types": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "items": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": true, "fields": {"actor": {"nullable": false, "required": true, "schema": {"type": "string"}}, "actor_id": {"nullable": true, "required": true, "schema": {"type": "string"}}, "actor_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "after": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {"type": "object"}}, "at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "audit_id": {"nullable": false, "required": true, "schema": {"type": "integer"}}, "before": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {"type": "object"}}, "detail": {"nullable": true, "required": true, "schema": {"type": "string"}}, "entity": {"nullable": false, "required": true, "schema": {"type": "string"}}, "entity_id": {"nullable": false, "required": true, "schema": {"type": "string"}}, "entity_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "event": {"nullable": false, "required": true, "schema": {"type": "string"}}, "event_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "occurred_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "type": "object"}, "type": "array"}, "page": {"nullable": false, "required": true, "schema": {"type": "integer"}}, "size": {"nullable": false, "required": true, "schema": {"type": "integer"}}, "total": {"nullable": false, "required": true, "schema": {"type": "integer"}}, "type": "object", "shape": "object"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "array"}}, "type": "object"}, "shape": "object"}}
```

35 GET /health

RULES []

```
{ "request": { "body": null, "header": {}, "path": {}, "query": {}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"status": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "shape": "object"}}
```

36 GET /health/ready

RULES ["all checks PASS yields status READY and HTTP 200", "any check FAIL yields status NOT_READY and HTTP 503"]

```
{ "request": { "body": null, "header": {}, "path": {}, "query": {}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"checks": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"kafka": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "reason_code": {"nullable": true, "required": true, "schema": {"enum": ["CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["FAIL", "PASS"], "type": "string"}}, "type": "object"}}, "neo4j": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "reason_code": {"nullable": true, "required": true, "schema": {"enum": ["CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["FAIL", "PASS"], "type": "string"}}, "type": "object"}}, "postgresql_runtime": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "reason_code": {"nullable": true, "required": true, "schema": {"enum": ["CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["FAIL", "PASS"], "type": "string"}}, "type": "object"}}, "rag":
```

```
{
  "nullable": false, "required": true, "schema": {
    "additional_properties": false, "fields": {
      "latency_ms": {
        "nullable": false, "required": true, "schema": {
          "minimum": 0.0, "type": "integer"
        }
      }, "reason_code": {
        "enum": [
          "CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"
        ], "type": "string"
      }, "status": {
        "enum": [
          "FAIL", "PASS"
        ], "type": "string"
      }
    }, "type": "object"
  }, "reference_migration": {
    "nullable": false, "required": true, "schema": {
      "additional_properties": false, "fields": {
        "latency_ms": {
          "nullable": false, "required": true, "schema": {
            "minimum": 0.0, "type": "integer"
          }
        }, "reason_code": {
          "enum": [
            "CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"
          ], "type": "string"
        }, "status": {
          "enum": [
            "FAIL", "PASS"
          ], "type": "string"
        }
      }, "type": "object"
    }, "dataset_epoch": {
      "nullable": false, "required": true, "schema": {
        "minimum": 0.0, "type": "integer"
      }, "reason_code": {
        "enum": [
          "CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"
        ], "type": "string"
      }, "status": {
        "enum": [
          "FAIL", "PASS"
        ], "type": "string"
      }
    }, "shape": {
      "object": {}
    }, "503": {
      "schema": {
        "additional_properties": false, "fields": {
          "checks": {
            "nullable": false, "required": true, "schema": {
              "additional_properties": false, "fields": {
                "kafka": {
                  "nullable": false, "required": true, "schema": {
                    "additional_properties": false, "fields": {
                      "latency_ms": {
                        "nullable": false, "required": true, "schema": {
                          "minimum": 0.0, "type": "integer"
                        }
                      }, "reason_code": {
                        "enum": [
                          "CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"
                        ], "type": "string"
                      }, "status": {
                        "enum": [
                          "FAIL", "PASS"
                        ], "type": "string"
                      }
                    }, "type": "object"
                  }, "n8n": {
                    "nullable": false, "required": true, "schema": {
                      "additional_properties": false, "fields": {
                        "latency_ms": {
                          "nullable": false, "required": true, "schema": {
                            "minimum": 0.0, "type": "integer"
                          }
                        }, "reason_code": {
                          "enum": [
                            "CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"
                          ], "type": "string"
                        }, "status": {
                          "enum": [
                            "FAIL", "PASS"
                          ], "type": "string"
                        }
                      }, "type": "object"
                    }, "postgres_runtime": {
                      "nullable": false, "required": true, "schema": {
                        "additional_properties": false, "fields": {
                          "latency_ms": {
                            "nullable": false, "required": true, "schema": {
                              "minimum": 0.0, "type": "integer"
                            }
                          }, "reason_code": {
                            "enum": [
                              "CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"
                            ], "type": "string"
                          }, "status": {
                            "enum": [
                              "FAIL", "PASS"
                            ], "type": "string"
                          }
                        }, "type": "object"
                      }, "neo4j": {
                        "nullable": false, "required": true, "schema": {
                          "additional_properties": false, "fields": {
                            "latency_ms": {
                              "nullable": false, "required": true, "schema": {
                                "minimum": 0.0, "type": "integer"
                              }
                            }, "reason_code": {
                              "enum": [
                                "CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"
                              ], "type": "string"
                            }, "status": {
                              "enum": [
                                "FAIL", "PASS"
                              ], "type": "string"
                            }
                          }, "type": "object"
                        }, "rag": {
                          "nullable": false, "required": true, "schema": {
                            "additional_properties": false, "fields": {
                              "latency_ms": {
                                "nullable": false, "required": true, "schema": {
                                  "minimum": 0.0, "type": "integer"
                                }
                              }, "reason_code": {
                                "enum": [
                                  "CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"
                                ], "type": "string"
                              }, "status": {
                                "enum": [
                                  "FAIL", "PASS"
                                ], "type": "string"
                              }
                            }, "type": "object"
                          }, "reference_migration": {
                            "nullable": false, "required": true, "schema": {
                              "additional_properties": false, "fields": {
                                "latency_ms": {
                                  "nullable": false, "required": true, "schema": {
                                    "minimum": 0.0, "type": "integer"
                                  }
                                }, "reason_code": {
                                  "enum": [
                                    "CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"
                                  ], "type": "string"
                                }, "status": {
                                  "enum": [
                                    "FAIL", "PASS"
                                  ], "type": "string"
                                }
                              }, "type": "object"
                            }, "dataset_epoch": {
                              "nullable": false, "required": true, "schema": {
                                "minimum": 0.0, "type": "integer"
                              }, "reason_code": {
                                "enum": [
                                  "CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"
                                ], "type": "string"
                              }, "status": {
                                "enum": [
                                  "FAIL", "PASS"
                                ], "type": "string"
                              }
                            }, "shape": {
                              "object": {}
                            }
                          }
                        }
                      }
                    }
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}, "dto_components": {
  "action_code": {
    "enum": [
      "EQP_HOLD", "MONITORING", "WARNING"
    ], "type": "string"
  }, "action_delivery_detail_item": {
    "additional_properties": false, "fields": {
      "channel": {
        "nullable": false, "required": true, "schema": {
          "enum": [
            "EMAIL", "MES"
          ], "type": "string"
        }
      }, "completed_at": {
        "nullable": true, "required": true, "schema": {
          "format": "date-time", "type": "string"
        }
      }, "started_at": {
        "nullable": true, "required": true, "schema": {
          "format": "date-time", "type": "string"
        }
      }, "status": {
        "enum": [
          "BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"
        ], "type": "string"
      }
    }, "type": "object"
  }, "action_delivery_item": {
    "additional_properties": false, "fields": {
      "channel": {
        "nullable": false, "required": true, "schema": {
          "enum": [
            "EMAIL", "MES"
          ], "type": "string"
        }
      }, "status": {
        "enum": [
          "BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"
        ], "type": "string"
      }
    }, "type": "object"
  }, "action_detail_response": {
    "additional_properties": false, "fields": {
      "action_code": {
        "nullable": false, "required": true, "schema": {
          "enum": [
            "EQP_HOLD", "MONITORING", "WARNING"
          ], "type": "string"
        }
      }, "action_id": {
        "nullable": false, "required": true, "schema": {
          "min_length": 1, "type": "string"
        }
      }, "agent_run_id": {
        "nullable": false, "required": true, "schema": {
          "min_length": 1, "type": "string"
        }
      }, "approval_status": {
        "enum": [
          "APPROVED", "PENDING", "REJECTED"
        ], "type": "string"
      }, "chamber": {
        "nullable": false, "required": true, "schema": {
          "min_length": 1, "type": "string"
        }
      }, "created_at": {
        "nullable": false, "required": true, "schema": {
          "format": "date-time", "type": "string"
        }
      }, "created_by_agent_run_id": {
        "nullable": false, "required": true, "schema": {
          "min_length": 1, "type": "string"
        }
      }, "deliveries": {
        "nullable": false, "required": true, "schema": {
          "items": {
            "additional_properties": false, "fields": {
              "channel": {
                "enum": [
                  "EMAIL", "MES"
                ], "type": "string"
              }, "completed_at": {
                "nullable": true, "required": true, "schema": {
                  "format": "date-time", "type": "string"
                }
              }, "started_at": {
                "nullable": true, "required": true, "schema": {
                  "format": "date-time", "type": "string"
                }
              }, "status": {
                "enum": [
                  "BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"
                ], "type": "string"
              }
            }, "type": "object"
          }
        }
      }
    }
  }
}
}
```



```
: false, "required": true, "schema": {"enum": ["TRACE"], "type": "string"}}, "type": "object"}}, "type":
: "array"}}, {"limit": {"nullable": false, "required": true, "schema": {"type": "string"}}, "limitations":
{"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "predic
ted_fault_code": {"nullable": true, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TM
D"], "type": "string"}}, "recommended_action": {"nullable": true, "required": true, "schema": {"enum": ["
EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, {"title": {"nullable": false, "required": true, "
schema": {"min_length": 1, "type": "string"}}, {"tools": {"nullable": false, "required": true, "schema": {
"items": {"additional_properties": false, "fields": {"name": {"nullable": false, "required": true, "schem
a": {"min_length": 1, "type": "string"}}, "result": {"nullable": false, "required": true, "schema": {"min
_length": 1, "type": "string"}}, {"result_summary": {"nullable": false, "required": true, "schema": {"min
_length": 1, "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["ERRO
R", "SUCCESS", "TIMEOUT"], "type": "string"}}, {"tool_name": {"nullable": false, "required": true, "schema
": {"min_length": 1, "type": "string"}}, {"type": "object"}, {"type": "array"}}, {"rules": [{"fault_code an
d ground_truth_fault_code are forbidden"], "type": "object"}
AgentEvaluationResponse
{"additional_properties": false, "fields": {"fault_5class": {"nullable": true, "required": true, "schema"
: {"additional_properties": false, "fields": {"classification": {"nullable": false, "required": true, "sc
hema": {"additional_properties": false, "fields": {"accuracy": {"nullable": false, "required": true, "sch
ema": {"additional_properties": false, "fields": {"denominator": {"nullable": false, "required": true, "s
chema": {"minimum": 0.0, "type": "integer"}}, "numerator": {"nullable": false, "required": true, "schema"
: {"minimum": 0.0, "type": "integer"}}, "rate": {"nullable": false, "required": true, "schema": {"maximum
": 1.0, "minimum": 0.0, "type": "number"}}, {"type": "object"}}, {"by_class": {"nullable": false, "require
d": true, "schema": {"additional_properties": {"additional_properties": false, "fields": {"f1": {"nullabl
e": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, {"false_negati
ve": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, {"false_positiv
e": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, {"precision": {"
nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, {"recal
l": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}},
"support": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, {"true_po
sitive": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, {"type": "
object"}, {"fields": {}, {"type": "object"}}, {"macro_f1_5class": {"nullable": false, "required": true, "sch
ema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, {"observed_class_macro_f1": {"nullable": false,
"required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, {"population_count": {"n
ullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, {"unclassified_count":
{"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, {"type": "object"}},
{"evidence_valid_run": {"nullable": false, "required": true, "schema": {"additional_properties": false,
"fields": {"denominator": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integ
er"}}, {"numerator": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}},
"rate": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number
"}}, {"type": "object"}}, {"exclusions": {"nullable": false, "required": true, "schema": {"items": {"addit
ional_properties": false, "fields": {"count": {"nullable": false, "required": true, "schema": {"minimum":
0.0, "type": "integer"}}, {"meaning": {"nullable": false, "required": true, "schema": {"min_length": 1, "
type": "string"}}, {"reason": {"nullable": false, "required": true, "schema": {"enum": ["AMBIGUOUS_LABEL",
"NO_INJECTED_FAULT"], "type": "string"}}, {"type": "object"}, {"type": "array"}}, {"hard_gate_passed": {"n
ullable": false, "required": true, "schema": {"type": "boolean"}}, {"hard_gate_reasons": {"nullable": fals
e, "required": true, "schema": {"type": "string"}, {"type": "array"}}, {"label_source": {"nullabl
e": false, "required": true, "schema": {"type": "string"}}, {"metrology_observed_count": {"nullable": fals
e, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, {"metrology_total_lot_hist_count": {"
nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, {"production_ground_tr
uth_available": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, {"production_perform
ance_disclaimer": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"
public_fault_ground_truth_available": {"nullable": false, "required": true, "schema": {"type": "boolean"}
}, {"rule_action_agreement": {"nullable": false, "required": true, "schema": {"additional_properties": fal
se, "fields": {"denominator": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "i
nteger"}}, {"numerator": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer
"}}, {"rate": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "nu
mber"}}, {"type": "object"}}, {"structured_prediction": {"nullable": false, "required": true, "schema": {"
additional_properties": false, "fields": {"denominator": {"nullable": false, "required": true, "schema":
{"minimum": 0.0, "type": "integer"}}, {"numerator": {"nullable": false, "required": true, "schema": {"mini
mum": 0.0, "type": "integer"}}, {"rate": {"nullable": false, "required": true, "schema": {"maximum": 1.0,
"minimum": 0.0, "type": "number"}}, {"type": "object"}}, {"usage_scope": {"nullable": false, "required": t
rue, "schema": {"type": "string"}}, {"versions": {"nullable": false, "required": true, "schema": {"additio
nal_properties": false, "fields": {"dataset_epoch": {"nullable": false, "required": true, "schema": {"min
_length": 1, "type": "string"}}, {"model_version": {"nullable": true, "required": true, "schema": {"type":
"string"}}, {"policy_version": {"nullable": true, "required": true, "schema": {"type": "string"}}, {"promp
t_version": {"nullable": true, "required": true, "schema": {"type": "string"}}, {"type": "object"}}, {"ty
pe": "object"}}, {"fault_5class_empty_reason": {"nullable": true, "required": true, "schema": {"enum": ["C
ONTRACT_VIOLATION", "NOT_CONFIGURED", "NOT_FOUND"], "type": "string"}}, {"golden_flow": {"nullable": true,
"required": true, "schema": {"additional_properties": false, "fields": {"dataset_epoch": {"nullable": fa
lse, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"phases": {"nullable": false, "req
uired": true, "schema": {"items": {"additional_properties": false, "fields": {"metrics": {"nullable": fal
se, "required": true, "schema": {"additional_properties": true, "fields": {}, {"type": "object"}}, {"phase"
: {"nullable": false, "required": true, "schema": {"enum": ["BATCH_BASELINE", "DECISIONS", "MANUAL_RETRY"
, "PREFLIGHT", "PRE_APPROVAL", "SECOND_BATCH", "UNKNOWN"], "type": "string"}}, {"reasons": {"nullable": fa
```

```
lse, "required": true, "schema": {"items": {"type": "string", "type": "array"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["EVIDENCE_INCOMPLETE", "FAIL", "PASS"], "type": "string"}}, "type": "object"}, "type": "array"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["EVIDENCE_INCOMPLETE", "FAIL", "PASS"], "type": "string"}}, "type": "object"}}, "golden_flow_empty_reason": {"nullable": true, "required": true, "schema": {"enum": ["CONTRACT_VIOLATION", "NOT_CONFIGURED", "NOT_FOUND"], "type": "string"}}, "type": "object"}
AgentPredictionDetailItem
{"additional_properties": false, "fields": {"cause_summary": {"nullable": false, "required": true, "schema": {"max_length": 2000, "min_length": 1, "type": "string"}}, "confidence": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "generated_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "input_tokens": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "lm_model": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "output_tokens": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "predicted_fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "prompt_version": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "supporting_alarms": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "type": "object"}, "type": "array"}}, "supporting_chunk_ids": {"nullable": false, "required": true, "schema": {"items": {"min_length": 1, "type": "string"}, "type": "array"}}, "supporting_relation_ids": {"nullable": false, "required": true, "schema": {"items": {"min_length": 1, "type": "string"}, "type": "array"}}, "uncertainty": {"nullable": false, "required": true, "schema": {"max_length": 1000, "type": "string"}}, "type": "object"}
AgentRunAccepted
{"additional_properties": false, "fields": {"agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "alarm": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "type": "object"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["RUNNING"], "type": "string"}}, "type": "object"}
AgentRunAcceptedResponse
{"additional_properties": false, "fields": {"agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "alarm": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "type": "object"}}, "status": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}
AgentRunActionItem
{"additional_properties": false, "fields": {"action_code": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_status": {"nullable": true, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "deliveries": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES"], "type": "string"}}, "completed_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "started_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, "type": "object"}, "type": "array"}}, "reason": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}
AgentRunApprovalItem
{"additional_properties": false, "fields": {"action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "decided_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_by": {"nullable": true, "required": true, "schema": {"type": "string"}}, "decision_comment": {"nullable": true, "required": true, "schema": {"type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "type": "object"}
AgentRunAskEvidence
{"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"max_length": 4000, "min_length": 1, "type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}
AgentRunCreateRequest
{"additional_properties": false, "fields": {"alarm": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "type": "object"}}, "type": "object"}
AgentRunDetailResponse
{"additional_properties": false, "fields": {"action": {"nullable": true, "required": true, "schema": {"additional_properties": false, "fields": {"action_code": {"nullable": false, "required": true, "schema": {"
```

```
enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_status": {"nullable": true, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "deliveries": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES"], "type": "string"}}, "complete_d_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "started_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, "type": "object"}, "reason": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}}, "action_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "alarm_source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "approval": {"nullable": true, "required": true, "schema": {"additional_properties": false, "fields": {"action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "decided_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_by": {"nullable": true, "required": true, "schema": {"type": "string"}}, "decision_comment": {"nullable": true, "required": true, "schema": {"type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "type": "object"}}, "approval_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "autonomy_level": {"nullable": false, "required": false, "schema": {"default": 2, "maximum": 3.0, "minimum": 1.0, "type": "integer"}}, "chamber": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "confidence": {"nullable": true, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "deliveries": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, "type": "object"}, "type": "array"}}, "diagnosis": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"alternative_hypotheses": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"lower_rank_reason": {"nullable": false, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "summary": {"nullable": false, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "cause_summary": {"nullable": true, "required": false, "schema": {"max_length": 2000, "type": "string"}}, "confidence": {"nullable": true, "required": false, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "diagnostic_coverage": {"nullable": true, "required": false, "schema": {"type": "string"}}, "evidence_synthesis": {"nullable": true, "required": false, "schema": {"max_length": 2000, "type": "string"}}, "limitations": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "observations": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "origin_assessment": {"nullable": true, "required": false, "schema": {"additional_properties": false, "fields": {"basis": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"id": {"nullable": false, "required": true, "schema": {"max_length": 160, "min_length": 1, "type": "string"}}, "namespace": {"nullable": false, "required": true, "schema": {"enum": ["ALARM", "CHUNK", "LOT_HIST", "PARAMETER", "RELATION"], "type": "string"}}, "type": "object"}, "type": "array"}}, "compared": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"downstream": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "history": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "metrology": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "sibling": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "upstream": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "type": "object"}}, "scope": {"nullable": false, "required": true, "schema": {"enum": ["CURRENT_CHAMBER", "DOWNSTREAM", "EQUIPMENT_COMMON", "UNDETERMINED", "UPSTREAM"], "type": "string"}}, "type": "object"}}, "parameter_findings": {"nullable": false, "required": false, "schema": {"default": [], "items": {"additional_properties": false, "fields": {"direction": {"nullable": false, "required": true, "schema": {"enum": ["ABOVE", "BELOW", "BOTH"], "type": "string"}}, "excursion_ratio": {"nullable": false, "required": true, "schema": {"type": "number"}}, "lot_hist_ids": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "parameter_id": {"nullable": false, "required": true, "schema": {"max_length": 20, "min_length": 1, "type": "string"}}, "step_no": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "wafer_scope": {"nullable": false, "required": true, "schema": {"enum": ["ALL", "PARTIAL", "SINGLE"], "type": "string"}}, "type": "object"}, "type": "array"}}, "predicted_fault_code": {"nullable": true, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "reason_code": {"nullable": true, "required": false, "schema": {"pattern": "^A-Z0-9_+$$", "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["AVAILABLE", "EMPTY"], "type": "string"}}, "verification_steps": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "type": "object"}}, "evidence_assessment": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"available_sources": {"nullab
```

le": false, "required": true, "schema": {"items": {"enum": ["FDC", "GRAPH", "POSTGRES_ROUTE", "RAG"], "type": "string"}, "type": "array"}}, "conflicting_source_ids": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "missing_sources": {"nullable": false, "required": true, "schema": {"items": {"enum": ["FDC", "GRAPH", "POSTGRES_ROUTE", "RAG"], "type": "string"}, "type": "array"}}, "reason_codes": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["CONFLICT", "EMPTY", "PARTIAL", "SUFFICIENT"], "type": "string"}}, "type": "object"}, "evidence_items": {"nullable": false, "required": true, "schema": {"items": {"discriminator": "type", "type": "discriminated_union", "variants": {"ALARM": {"additional_properties": false, "fields": {"alarm": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "type": "object"}, "excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "DOCUMENT": {"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "section": {"nullable": true, "required": true, "schema": {"type": "string"}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "METROLOGY": {"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "TRACE": {"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}}, "type": "array"}, "fault_code": {"nullable": true, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}, "fault_color": {"nullable": false, "required": true, "schema": {"type": "null"}, "fault_name": {"nullable": false, "required": true, "schema": {"type": "null"}, "impact_scope": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"check_required": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"kind": {"nullable": false, "required": true, "schema": {"enum": ["CHAMBER", "LOT", "PARAMETER", "PROCESS_STEP", "SIBLING_CHAMBER", "WAFER"], "type": "string"}, "relation": {"nullable": true, "required": false, "schema": {"type": "string"}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "type": {"object"}, "type": "array"}}, "graph_conflict": {"nullable": false, "required": false, "schema": {"default": false, "type": "boolean"}, "reason_code": {"nullable": true, "required": false, "schema": {"pattern": "[A-Z0-9_]+", "type": "string"}, "status": {"nullable": false, "required": true, "schema": {"enum": ["AVAILABLE", "EMPTY"], "type": "string"}, "summary": {"nullable": true, "required": false, "schema": {"max_length": 2000, "type": "string"}}, "type": "object"}, "latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}, "llm_model": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "post_action_observation": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"message": {"nullable": false, "required": false, "schema": {"default": "\u005c\u0085 \u0015\u001b \u0037\u0074\u0013\u0014\u005d\u0029 \u00c8\u0074\u0064 \u00f5\u0015 \u00d0\u0021\u0012\u0074 \u0056\u005b \u00d6\u00a8\u00ac\u00b9\u007c \u00d3\u00c9\u00d5\u0060 \u00218 \u0056\u0074c", "type": "string"}, "status": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE_STATIC_DATASET", "type": "string"}}, "type": "object"}, "predicted_fault_code": {"nullable": true, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}, "prediction": {"nullable": true, "required": true, "schema": {"additional_properties": false, "fields": {"cause_summary": {"nullable": false, "required": true, "schema": {"max_length": 2000, "min_length": 1, "type": "string"}, "confidence": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}, "generated_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}, "input_tokens": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}, "llm_model": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "output_tokens": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}, "predicted_fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}, "prompt_version": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "supporting_alarms": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"alarm_id": {"nulla

```

false, "required": true, "schema": {"min_length": 1, "type": "string"}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "type": "object"}, {"type": "array"}, {"supporting_chunk_ids": {"nullable": false, "required": true, "schema": {"items": {"min_length": 1, "type": "string"}, "type": "array"}}, {"supporting_relation_ids": {"nullable": false, "required": true, "schema": {"items": {"min_length": 1, "type": "string"}, "type": "array"}}, {"uncertainty": {"nullable": false, "required": true, "schema": {"max_length": 1000, "type": "string"}}, {"type": "object"}, {"react_trace": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": false, "fields": {"argument_summary": {"nullable": true, "required": false, "schema": {"pattern": "^(\\ud6c4\\ubcf4\\u0009\\u0020\\u002d\\u002e\\u002f\\u003a\\u003b\\u003c\\u003e\\u003f\\u0040\\u005b\\u005c\\u005d\\u005e\\u005f\\u0060\\u0061\\u0062\\u0063\\u0064\\u0065\\u0066\\u0067\\u0068\\u0069\\u006a\\u006b\\u006c\\u006d\\u006e\\u006f\\u0070\\u0071\\u0072\\u0073\\u0074\\u0075\\u0076\\u0077\\u0078\\u0079\\u007a\\u007b\\u007c\\u007d\\u007e\\u007f\\u0080\\u0081\\u0082\\u0083\\u0084\\u0085\\u0086\\u0087\\u0088\\u0089\\u008a\\u008b\\u008c\\u008d\\u008e\\u008f\\u0090\\u0091\\u0092\\u0093\\u0094\\u0095\\u0096\\u0097\\u0098\\u0099\\u00a0\\u00a1\\u00a2\\u00a3\\u00a4\\u00a5\\u00a6\\u00a7\\u00a8\\u00a9\\u00aa\\u00ab\\u00ac\\u00ad\\u00ae\\u00af\\u00b0\\u00b1\\u00b2\\u00b3\\u00b4\\u00b5\\u00b6\\u00b7\\u00b8\\u00b9\\u00c0\\u00c1\\u00c2\\u00c3\\u00c4\\u00c5\\u00c6\\u00c7\\u00c8\\u00c9\\u00ca\\u00cb\\u00cc\\u00cd\\u00ce\\u00cf\\u00d0\\u00d1\\u00d2\\u00d3\\u00d4\\u00d5\\u00d6\\u00d7\\u00d8\\u00d9\\u00da\\u00db\\u00dc\\u00dd\\u00de\\u00df\\u00e0\\u00e1\\u00e2\\u00e3\\u00e4\\u00e5\\u00e6\\u00e7\\u00e8\\u00e9\\u00f0\\u00f1\\u00f2\\u00f3\\u00f4\\u00f5\\u00f6\\u00f7\\u00f8\\u00f9\\u00fa\\u00fb\\u00fc\\u00fd\\u00fe\\u00ff\\u0100\\u0101\\u0102\\u0103\\u0104\\u0105\\u0106\\u0107\\u0108\\u0109\\u010a\\u010b\\u010c\\u010d\\u010e\\u010f\\u0110\\u0111\\u0112\\u0113\\u0114\\u0115\\u0116\\u0117\\u0118\\u0119\\u011a\\u011b\\u011c\\u011d\\u011e\\u011f\\u0120\\u0121\\u0122\\u0123\\u0124\\u0125\\u0126\\u0127\\u0128\\u0129\\u012a\\u012b\\u012c\\u012d\\u012e\\u012f\\u0130\\u0131\\u0132\\u0133\\u0134\\u0135\\u0136\\u0137\\u0138\\u0139\\u013a\\u013b\\u013c\\u013d\\u013e\\u013f\\u0140\\u0141\\u0142\\u0143\\u0144\\u0145\\u0146\\u0147\\u0148\\u0149\\u014a\\u014b\\u014c\\u014d\\u014e\\u014f\\u0150\\u0151\\u0152\\u0153\\u0154\\u0155\\u0156\\u0157\\u0158\\u0159\\u015a\\u015b\\u015c\\u015d\\u015e\\u015f\\u0160\\u0161\\u0162\\u0163\\u0164\\u0165\\u0166\\u0167\\u0168\\u0169\\u016a\\u016b\\u016c\\u016d\\u016e\\u016f\\u0170\\u0171\\u0172\\u0173\\u0174\\u0175\\u0176\\u0177\\u0178\\u0179\\u017a\\u017b\\u017c\\u017d\\u017e\\u017f\\u0180\\u0181\\u0182\\u0183\\u0184\\u0185\\u0186\\u0187\\u0188\\u0189\\u018a\\u018b\\u018c\\u018d\\u018e\\u018f\\u0190\\u0191\\u0192\\u0193\\u0194\\u0195\\u0196\\u0197\\u0198\\u0199\\u019a\\u019b\\u019c\\u019d\\u019e\\u019f\\u01a0\\u01a1\\u01a2\\u01a3\\u01a4\\u01a5\\u01a6\\u01a7\\u01a8\\u01a9\\u01aa\\u01ab\\u01ac\\u01ad\\u01ae\\u01af\\u01b0\\u01b1\\u01b2\\u01b3\\u01b4\\u01b5\\u01b6\\u01b7\\u01b8\\u01b9\\u01ba\\u01bb\\u01bc\\u01bd\\u01be\\u01bf\\u01c0\\u01c1\\u01c2\\u01c3\\u01c4\\u01c5\\u01c6\\u01c7\\u01c8\\u01c9\\u01ca\\u01cb\\u01cc\\u01cd\\u01ce\\u01cf\\u01d0\\u01d1\\u01d2\\u01d3\\u01d4\\u01d5\\u01d6\\u01d7\\u01d8\\u01d9\\u01da\\u01db\\u01dc\\u01dd\\u01de\\u01df\\u01e0\\u01e1\\u01e2\\u01e3\\u01e4\\u01e5\\u01e6\\u01e7\\u01e8\\u01e9\\u01ea\\u01eb\\u01ec\\u01ed\\u01ee\\u01ef\\u01f0\\u01f1\\u01f2\\u01f3\\u01f4\\u01f5\\u01f6\\u01f7\\u01f8\\u01f9\\u01fa\\u01fb\\u01fc\\u01fd\\u01fe\\u01ff\\u0200\\u0201\\u0202\\u0203\\u0204\\u0205\\u0206\\u0207\\u0208\\u0209\\u020a\\u020b\\u020c\\u020d\\u020e\\u020f\\u0210\\u0211\\u0212\\u0213\\u0214\\u0215\\u0216\\u0217\\u0218\\u0219\\u021a\\u021b\\u021c\\u021d\\u021e\\u021f\\u0220\\u0221\\u0222\\u0223\\u0224\\u0225\\u0226\\u0227\\u0228\\u0229\\u022a\\u022b\\u022c\\u022d\\u022e\\u022f\\u0230\\u0231\\u0232\\u0233\\u0234\\u0235\\u0236\\u0237\\u0238\\u0239\\u023a\\u023b\\u023c\\u023d\\u023e\\u023f\\u0240\\u0241\\u0242\\u0243\\u0244\\u0245\\u0246\\u0247\\u0248\\u0249\\u024a\\u024b\\u024c\\u024d\\u024e\\u024f\\u0250\\u0251\\u0252\\u0253\\u0254\\u0255\\u0256\\u0257\\u0258\\u0259\\u025a\\u025b\\u025c\\u025d\\u025e\\u025f\\u0260\\u0261\\u0262\\u0263\\u0264\\u0265\\u0266\\u0267\\u0268\\u0269\\u026a\\u026b\\u026c\\u026d\\u026e\\u026f\\u0270\\u0271\\u0272\\u0273\\u0274\\u0275\\u0276\\u0277\\u0278\\u0279\\u027a\\u027b\\u027c\\u027d\\u027e\\u027f\\u0280\\u0281\\u0282\\u0283\\u0284\\u0285\\u0286\\u0287\\u0288\\u0289\\u028a\\u028b\\u028c\\u028d\\u028e\\u028f\\u0290\\u0291\\u0292\\u0293\\u0294\\u0295\\u0296\\u0297\\u0298\\u0299\\u029a\\u029b\\u029c\\u029d\\u029e\\u029f\\u02a0\\u02a1\\u02a2\\u02a3\\u02a4\\u02a5\\u02a6\\u02a7\\u02a8\\u02a9\\u02aa\\u02ab\\u02ac\\u02ad\\u02ae\\u02af\\u02b0\\u02b1\\u02b2\\u02b3\\u02b4\\u02b5\\u02b6\\u02b7\\u02b8\\u02b9\\u02ba\\u02bb\\u02bc\\u02bd\\u02be\\u02bf\\u02c0\\u02c1\\u02c2\\u02c3\\u02c4\\u02c5\\u02c6\\u02c7\\u02c8\\u02c9\\u02ca\\u02cb\\u02cc\\u02cd\\u02ce\\u02cf\\u02d0\\u02d1\\u02d2\\u02d3\\u02d4\\u02d5\\u02d6\\u02d7\\u02d8\\u02d9\\u02da\\u02db\\u02dc\\u02dd\\u02de\\u02df\\u02e0\\u02e1\\u02e2\\u02e3\\u02e4\\u02e5\\u02e6\\u02e7\\u02e8\\u02e9\\u02ea\\u02eb\\u02ec\\u02ed\\u02ee\\u02ef\\u
```



```
rue, "schema": {"enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}}, "tools": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"n": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result_summary": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "s": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "tool_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"type": "object"}, "type": "array"}}, "type": "object"}

AlarmAskEvidence
{"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}

AlarmEvidence
{"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["ALARM"], "type": "string"}}, "type": "object"}

AlarmItem
{"additional_properties": false, "fields": {"action_code": {"nullable": true, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "alarm_type": {"nullable": false, "required": true, "schema": {"enum": ["OOC", "OOS"], "type": "string"}}, "area": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "cl": {"nullable": true, "required": true, "schema": {"type": "number"}}, "equipment": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "fault": {"nullable": true, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "icl": {"nullable": true, "required": true, "schema": {"type": "number"}}, "lot": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "mes": {"nullable": false, "required": true, "schema": {"type": "string"}}, "mes_status": {"nullable": true, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, "notify": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "notify_status": {"nullable": true, "required": true, "schema": {"enum": ["FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, "occurred_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "parameter": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "parameter_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "predicted_fault_code": {"nullable": true, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "recipe": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "recipe_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "recipe_step_no": {"nullable": false, "required": true, "schema": {"minimum": 1, "type": "integer"}}, "rule_code": {"nullable": false, "required": true, "schema": {"enum": ["R03_CONSEC", "SUMMARY_OOC", "TRACE_OOS"], "type": "string"}}, "seq_no": {"nullable": true, "required": true, "schema": {"minimum": 0, "type": "integer"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "statistic_type": {"nullable": true, "required": true, "schema": {"type": "string"}}, "step_no": {"nullable": false, "required": true, "schema": {"minimum": 1, "type": "integer"}}, "ucl": {"nullable": true, "required": true, "schema": {"type": "number"}}, "value": {"nullable": true, "required": true, "schema": {"type": "number"}}, "wafer": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "wafer_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "rules": ["source TRACE requires rule_code TRACE_OOS", "source SUMMARY requires rule_code SUMMARY_OOC", "source R03 requires alarm_type OOS, rule_code R03_CONSEC, and value null", "compatibility aliases derive only from canonical fields"], "type": "object"}

AlarmRef
{"additional_properties": false, "fields": {"alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "type": "object"}

AlarmSource
{"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}

AnalysisQueryRequest
{"additional_properties": false, "fields": {"question": {"nullable": false, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "type": "object"}

AnalysisQueryResponse
{"additional_properties": false, "fields": {"columns": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "cross_check": {"nullable": true, "required": false, "schema": {"additional_properties": false, "fields": {"cypher": {"nullable": true, "required": false, "schema": {"type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["MATCH", "MISMATCH", "SKIPPED"], "type": "string"}}, "summary": {"nullable": true, "required": false, "schema": {"type": "string"}}, "type": "object"}}, "error_msg": {"nullable": true, "required": false, "schema": {"type": "string"}}, "generated_sql": {"nullable": true, "required": false, "schema": {"type": "string"}}, "group
```

```
by": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "is_rejected": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "is_valid": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "metric": {"nullable": true, "required": false, "schema": {"additional_properties": false, "fields": {"column": {"nullable": true, "required": false, "schema": {"type": "string"}}, "p": {"nullable": true, "required": false, "schema": {"maximum": 100.0, "minimum": 0.0, "type": "number"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["count", "max", "mean", "median", "min", "percentile", "ratio", "std", "sum"], "type": "string"}}}, "type": "object"}}, "metric_result": {"nullable": false, "required": false, "schema": {"type": "union", "variants": [{"type": "integer"}, {"type": "number"}, {"items": {"additional_properties": false, "fields": {"group": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "value": {"nullable": false, "required": false, "schema": {"type": "union", "variants": [{"type": "integer"}, {"type": "number"}, {"type": "null"}]}}, "type": "object"}, {"type": "array"}, {"type": "null"}]}}, "nl_query_log_id": {"nullable": true, "required": false, "schema": {"minimum": 1.0, "type": "integer"}}, "question": {"nullable": false, "required": true, "schema": {"type": "string"}}, "reject_reason": {"nullable": true, "required": false, "schema": {"type": "string"}}, "row_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "rows": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": true, "fields": {}, "type": "object"}, "type": "array"}}, "visualization": {"nullable": true, "required": false, "schema": {"additional_properties": false, "fields": {"chart_type": {"nullable": false, "required": true, "schema": {"enum": ["bar", "histogram", "line", "table"], "type": "string"}}, "x": {"nullable": true, "required": false, "schema": {"type": "string"}}, "y": {"nullable": true, "required": false, "schema": {"type": "string"}}, "type": "object"}}, "type": "object"}
ApprovalDecisionRequest
{"additional_properties": false, "fields": {"decided_by": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "decision": {"nullable": false, "required": true, "schema": {"enum": ["APPROVED", "REJECTED"], "type": "string"}}, "decision_comment": {"nullable": true, "required": false, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "type": "object"}
ApprovalItem
{"additional_properties": false, "fields": {"action_code": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approved_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "approved_by": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_by": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "decision_comment": {"nullable": true, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "equipment": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "lot": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "predicted_fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "reason": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "rules": ["PENDING requires all decision fields null", "APPROVED or REJECTED requires decided_by and decided_at", "only EQP_HOLD actions have approvals"], "type": "object"}
AskDocumentEvidenceAlias
{"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "doc_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "section": {"nullable": true, "required": true, "schema": {"type": "string"}}, "type": "object"}
AskToolItem
{"additional_properties": false, "fields": {"name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result_summary": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "tool_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}
AuditLogItem
{"additional_properties": false, "fields": {"actor": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "actor_id": {"nullable": true, "required": true, "schema": {"type": "string"}}, "actor_type": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "after": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "audit_id": {"nullable": false, "required": true, "schema": {"minimum": 1, "type": "integer"}}, "before": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "detail": {"nullable": true, "required": true, "schema": {"type": "string"}}, "ent
```

```
ity": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "entity_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "entity_type": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "event": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "event_type": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "occurred_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "type": "object"}
```

AuditLogPageResponse

```
{"additional_properties": true, "fields": {"event_type_counts": {"nullable": false, "required": true, "schema": {"additional_properties": {"type": "integer"}, "fields": {}, "type": "object"}}, "event_types": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "items": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": true, "fields": {"actor": {"nullable": false, "required": true, "schema": {"type": "string"}}, "actor_id": {"nullable": true, "required": true, "schema": {"type": "string"}}, "actor_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "after": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "audit_id": {"nullable": false, "required": true, "schema": {"type": "integer"}}, "before": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {"type": "object"}}, "detail": {"nullable": true, "required": true, "schema": {"type": "string"}}, "entity": {"nullable": false, "required": true, "schema": {"type": "string"}}, "entity_id": {"nullable": false, "required": true, "schema": {"type": "string"}}, "entity_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "event": {"nullable": false, "required": true, "schema": {"type": "string"}}, "event_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "occurred_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "type": "object"}, "page": {"nullable": false, "required": true, "schema": {"type": "integer"}}, "size": {"nullable": false, "required": true, "schema": {"type": "integer"}}, "total": {"nullable": false, "required": true, "schema": {"type": "integer"}}, "type": "object"}
```

AutoToolCallItem

```
{"additional_properties": false, "fields": {"n": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result_summary": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "s": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "tool_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}
```

ChamberGraphResponse

```
{"additional_properties": false, "fields": {"context": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"adjacent_process_step_ids": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}, "area": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "model_code": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "parameter_ids": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "process_step_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "relation_ids": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "graph_revision": {"nullable": false, "required": true, "schema": {"pattern": "[0-9a-f]{64}$", "type": "string"}}, "node_count": {"nullable": false, "required": true, "schema": {"minimum": 0, "type": "integer"}}, "nodes": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"business_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "label": {"nullable": false, "required": true, "schema": {"enum": ["Area", "Chamber", "Equipment", "EquipmentModel", "Parameter", "ProcessStep"], "type": "string"}, "name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "node_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "properties": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "type": "object"}, "type": "array"}}, "relationship_count": {"nullable": false, "required": true, "schema": {"minimum": 0, "type": "integer"}}, "relationships": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"from_node_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "relation_id": {"nullable": false, "required": true, "schema": {"pattern": "^REL-[0-9a-f]{20}$", "type": "string"}}, "to_node_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "rules": ["node_count equals len(nodes)", "relationship_count equals len(relationships)"], "type": "object"}
```

ChamberRelationResponse

```
{"additional_properties": false, "fields": {"graph_revision": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "nodes": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"business_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "display_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "label": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "properties": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "type": "object"}, "type": "array"}}, "relationships": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "target": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "rules": ["node_count equals len(nodes)", "relationship_count equals len(relationships)"], "type": "object"}
```

```
rue, "schema": {"min_length": 1, "type": "string"}}, {"type": "object"}, {"type": "array"}}, {"root_node_id": 1, "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"type": "object"}]
}
ChartType
{"enum": ["bar", "histogram", "line", "table"], "type": "string"}
ChatToolCallItem
{"additional_properties": false, "fields": {"name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result_summary": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "tool_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"type": "object"}
ComparisonMatrix
{"additional_properties": false, "fields": {"downstream": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "history": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "metrology": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "sibling": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "upstream": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, {"type": "object"}
CrossCheck
{"additional_properties": false, "fields": {"cypher": {"nullable": true, "required": false, "schema": {"type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["MATCH", "MISMATCH", "SKIPPED"], "type": "string"}}, "summary": {"nullable": true, "required": false, "schema": {"type": "string"}}, {"type": "object"}
DeliveryCallbackRequest
{"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES MOCK"], "type": "string"}}, "completed_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "error_code": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "event_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "provider_message_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "request_hash": {"nullable": false, "required": true, "schema": {"max_length": 64, "min_length": 64, "pattern": "^[0-9a-f]{64}$", "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["FAILED", "SENT"], "type": "string"}}, "rules": ["SENT requires provider_message_id non-null and error_code null", "FAILED requires error_code non-null and permits provider_message_id null"], "type": "object"}
DeliveryChannel
{"enum": ["EMAIL", "MES MOCK"], "type": "string"}
DeliveryResult
{"additional_properties": false, "fields": {"action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES MOCK"], "type": "string"}}, "completed_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "duplicate": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "error_code": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "provider_message_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "request_hash": {"nullable": false, "required": true, "schema": {"max_length": 64, "min_length": 64, "pattern": "^[0-9a-f]{64}$", "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["FAILED", "SENT"], "type": "string"}}, "rules": ["SENT requires provider_message_id non-null and error_code null", "FAILED requires error_code non-null and permits provider_message_id null"], "type": "object"}
DeliveryStatus
{"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}
DiagnosisBlock
{"additional_properties": false, "fields": {"alternative_hypotheses": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"lower_rank_reason": {"nullable": false, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "summary": {"nullable": false, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, {"type": "object"}}, {"type": "array"}}, "cause_summary": {"nullable": true, "required": false, "schema": {"max_length": 2000, "type": "string"}}, "confidence": {"nullable": true, "required": false, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "diagnostic_coverage": {"nullable": true, "required": false, "schema": {"type": "string"}}, "evidence_synthesis": {"nullable": true, "required": false, "schema": {"max_length": 2000, "type": "string"}}, "limitations": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, {"type": "array"}}, "observations": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, {"type": "array"}}, "origin_assessment": {"nullable": true, "required": false, "schema": {"additional_properties": false, "fields": {"basis": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"id": {"nullable": false, "required": true, "schema": {"max_length": 160, "min_length": 1, "type": "string"}}, "namespace": {"nullable": false, "required": true, "schema": {"enum": ["ALARM", "CHUNK", "LOT_HIST", "PARAMETER", "RELATION"], "type": "string"}}, {"type": "object"}}, {"type": "array"}}, "compared": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"downstream": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "history": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CH
```

```
ECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "metrology": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "sibling": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "upstream": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "scope": {"nullable": false, "required": true, "schema": {"enum": ["CURRENT_CHAMBER", "DOWNSTREAM", "EQUIPMENT_COMMON", "UNDETERMINED", "UPSTREAM"], "type": "string"}}, "parameter_findings": {"nullable": false, "required": false, "schema": {"default": [], "items": {"additional_properties": false, "fields": {"direction": {"nullable": false, "required": true, "schema": {"enum": ["ABOVE", "BELOW", "BOTH"], "type": "string"}}, "excursion_ratio": {"nullable": false, "required": true, "schema": {"type": "number"}}, "lot_hist_ids": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "parameter_id": {"nullable": false, "required": true, "schema": {"max_length": 20, "min_length": 1, "type": "string"}}, "step_no": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "wafer_scope": {"nullable": false, "required": true, "schema": {"enum": ["ALL", "PARTIAL", "SINGLE"], "type": "string"}}, "type": "object"}, "type": "array"}}, "predicted_fault_code": {"nullable": true, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "reason_code": {"nullable": true, "required": false, "schema": {"pattern": "[A-Z0-9_]+$", "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["AVAILABLE", "EMPTY"], "type": "string"}}, "verification_steps": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "type": "object"}

DocumentAskEvidence
{"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "section": {"nullable": true, "required": true, "schema": {"type": "string"}}, "source_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}

DocumentChunkItem
{"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chunk_seq": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "content": {"nullable": false, "required": true, "schema": {"type": "string"}}, "section_title": {"nullable": true, "required": false, "schema": {"type": "string"}}, "type": "object"}

DocumentDetailResponse
{"additional_properties": false, "fields": {"chunks": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chunk_seq": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "content": {"nullable": false, "required": true, "schema": {"type": "string"}}, "section_title": {"nullable": true, "required": false, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "doc_type": {"nullable": true, "required": false, "schema": {"enum": ["MANUAL", "SPEC", "TROUBLESHOOT"], "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "model_code": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "source_path": {"nullable": true, "required": false, "schema": {"type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"type": "string"}}, "version": {"nullable": true, "required": false, "schema": {"type": "string"}}, "type": "object"}

DocumentEvidence
{"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "excerpt": {"nullable": false, "required": true, "schema": {"type": "string"}}, "section": {"nullable": true, "required": true, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["DOCUMENT"], "type": "string"}}, "type": "object"}

DocumentHit
{"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"pattern": "^\\.cs2-[0-9]{4}$", "type": "string"}}, "content": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "doc_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "model_code": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "score": {"nullable": false, "required": true, "schema": {"maximum": 1, "minimum": -1, "type": "number"}}, "section": {"nullable": true, "required": true, "schema": {"type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}

DocumentSearchRequest
{"additional_properties": false, "fields": {"doc_type": {"nullable": true, "required": false, "schema": {"enum": ["MANUAL", "SPEC", "TROUBLESHOOT"], "type": "string"}}, "model_code": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "query": {"nullable": false, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "top_k": {"nullable": false, "required": false, "schema": {"default": 4, "maximum": 10, "minimum": 1, "type": "integer"}}, "type": "object"}

DocumentType
{"enum": ["MANUAL", "SPEC", "TROUBLESHOOT"], "type": "string"}

ErrorCode
{"enum": ["APPROVAL_ALREADY_DECIDED", "DEPENDENCY_NOT_READY", "IDEMPOTENCY_CONFLICT", "INCIDENT_ALREADY_P
```

ROCESSED", "INCIDENT_ALREADY_RUNNING", "INTERNAL_ERROR", "LEGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_NOT_FOUND", "UNAUTHORIZED", "VALIDATION_ERROR", "type": "string"]

ErrorResponse

{ "additional_properties": false, "fields": { "code": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "details": { "nullable": false, "required": true, "schema": { "additional_properties": true, "fields": {}, "type": "object" } }, "message": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "type": "object" }

EvaluationListResponse

{ "additional_properties": false, "fields": { "items": { "nullable": false, "required": true, "schema": { "items": { "additional_properties": false, "fields": { "accuracy": { "nullable": false, "required": true, "schema": { "maximum": 1.0, "minimum": 0.0, "type": "number" }, "correct": { "nullable": false, "required": true, "schema": { "minimum": 0, "type": "integer" }, "defense_passed": { "nullable": false, "required": true, "schema": { "minimum": 0, "type": "integer" }, "defense_total": { "nullable": false, "required": true, "schema": { "minimum": 0, "type": "integer" }, "executed_at": { "nullable": false, "required": true, "schema": { "format": "date-time", "type": "string" }, "items": { "nullable": false, "required": true, "schema": { "items": { "additional_properties": false, "fields": { "actual_result": { "nullable": true, "required": false, "schema": { "type": "any" }, "actual_visualization": { "nullable": true, "required": false, "schema": { "additional_properties": false, "fields": { "chart_type": { "nullable": false, "required": true, "schema": { "enum": ["bar", "histogram", "line", "table"], "type": "string" }, "x": { "nullable": true, "required": false, "schema": { "type": "string" }, "y": { "nullable": true, "required": false, "schema": { "type": "string" } } }, "type": "object" }, "attempt_count": { "nullable": false, "required": true, "schema": { "minimum": 0, "type": "integer" }, "case_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "case_type": { "nullable": false, "required": true, "schema": { "enum": ["DEFENSE", "GOLD"], "type": "string" }, "expected_result": { "nullable": true, "required": false, "schema": { "type": "any" }, "expected_visualization": { "nullable": true, "required": false, "schema": { "additional_properties": false, "fields": { "chart_type": { "nullable": false, "required": true, "schema": { "enum": ["bar", "histogram", "line", "table"], "type": "string" }, "x": { "nullable": true, "required": false, "schema": { "type": "string" }, "y": { "nullable": true, "required": false, "schema": { "type": "string" } } }, "type": "object" }, "generated_sql": { "nullable": true, "required": false, "schema": { "type": "string" }, "latency_ms": { "nullable": true, "required": false, "schema": { "minimum": 0, "type": "integer" }, "passed": { "nullable": false, "required": true, "schema": { "type": "boolean" }, "question": { "nullable": true, "required": false, "schema": { "type": "string" }, "reason": { "nullable": true, "required": false, "schema": { "type": "string" } }, "type": "object" }, "type": "array" }, "model": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "prompt_version": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "provider": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "run_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "temperature": { "nullable": false, "required": true, "schema": { "type": "number" }, "total": { "nullable": false, "required": true, "schema": { "minimum": 0, "type": "integer" } }, "type": "object" }, "type": "array" }, "page": { "nullable": false, "required": true, "schema": { "minimum": 1, "type": "integer" }, "size": { "nullable": false, "required": true, "schema": { "maximum": 100, "minimum": 1, "type": "integer" }, "total": { "nullable": false, "required": true, "schema": { "minimum": 0, "type": "integer" } }, "type": "object" }

EvidenceAssessmentBlock

{ "additional_properties": false, "fields": { "available_sources": { "nullable": false, "required": true, "schema": { "items": { "enum": ["FDC", "GRAPH", "POSTGRES_ROUTE", "RAG"], "type": "string" }, "type": "array" }, "conflicting_source_ids": { "nullable": false, "required": true, "schema": { "items": { "type": "string" }, "type": "array" }, "missing_sources": { "nullable": false, "required": true, "schema": { "items": { "enum": ["FDC", "GRAPH", "POSTGRES_ROUTE", "RAG"], "type": "string" }, "type": "array" }, "reason_codes": { "nullable": false, "required": true, "schema": { "items": { "type": "string" }, "type": "array" }, "status": { "nullable": false, "required": true, "schema": { "enum": ["CONFLICT", "EMPTY", "PARTIAL", "SUFFICIENT"], "type": "string" } }, "type": "object" }

Evidenceltem

{ "discriminator": { "type": "discriminated_union", "variants": { "ALARM": { "additional_properties": false, "fields": { "excerpt": { "nullable": false, "required": true, "schema": { "type": "string" }, "source_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "title": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "type": { "nullable": false, "required": true, "schema": { "enum": ["ALARM"], "type": "string" } }, "type": "object" }, "DOCUMENT": { "additional_properties": false, "fields": { "chunk_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "document_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "excerpt": { "nullable": false, "required": true, "schema": { "type": "string" }, "section": { "nullable": true, "required": true, "schema": { "type": "string" }, "source_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "title": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "type": { "nullable": false, "required": true, "schema": { "enum": ["DOCUMENT"], "type": "string" } }, "type": "object" }, "GRAPH": { "additional_properties": false, "fields": { "excerpt": { "nullable": false, "required": true, "schema": { "type": "string" }, "graph_revision": { "nullable": false, "required": true, "schema": { "pattern": "[0-9a-f]{64}\$", "type": "string" }, "relation_id": { "nullable": false, "required": true, "schema": { "pattern": "^REL-[0-9a-f]{20}\$", "type": "string" }, "source_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "title": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "type": { "nullable": false, "required": true, "schema": { "enum": ["GRAPH"], "type": "string" } }, "type": "object" }, "METROLOGY": { "additional_properties": false, "fields": { "excerpt": { "nullable": false, "required": true, "schema": { "type": "string" }, "source_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "title": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "type": { "nullable": false, "required": true, "schema": { "enum": ["METROLOGY"], "type": "string" } }, "type": "object" } }, "type": "discriminated_union" }

```
a": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["METROLOGY"], "type": "string"}}, "rules": [{"metrology_alarm_result_is_forbidden": true, "type": "object"}], "TRACE": {"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["TRACE"], "type": "string"}}, "type": "object"}}}, "type": "object"}}
FaultHypothesis
{"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}
GraphAskEvidence
{"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "graph_revision": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "relation_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}
GraphContext
{"additional_properties": false, "fields": {"adjacent_process_step_ids": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "area": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "model_code": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "parameter_ids": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "process_step_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "relation_ids": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "type": "object"}
GraphEvidence
{"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"type": "string"}}, "graph_revision": {"nullable": false, "required": true, "schema": {"pattern": "\\0-9a-f{64}$", "type": "string"}}, "relation_id": {"nullable": false, "required": true, "schema": {"pattern": "\\REL-[0-9a-f]{20}$", "type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["GRAPH"], "type": "string"}}, "type": "object"}
GraphNode
{"additional_properties": false, "fields": {"business_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "label": {"nullable": false, "required": true, "schema": {"enum": ["Area", "Chamber", "Equipment", "EquipmentModel", "Parameter", "ProcessStep"], "type": "string"}}, "name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "node_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "properties": {"label": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "type": "object"}
GraphQueryRequest
{"additional_properties": false, "fields": {"question": {"nullable": false, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "type": "object"}
GraphQueryResponse
{"additional_properties": false, "fields": {"columns": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "error_msg": {"nullable": true, "required": false, "schema": {"type": "string"}}, "generated_cypher": {"nullable": true, "required": false, "schema": {"type": "string"}}, "is_rejected": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "is_valid": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "question": {"nullable": false, "required": true, "schema": {"type": "string"}}, "reject_reason": {"nullable": true, "required": false, "schema": {"type": "string"}}, "row_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "rows": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": true, "fields": {}, "type": "object"}, "type": "array"}}, "type": "object"}
GraphRelationship
{"additional_properties": false, "fields": {"from_node_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "relation_id": {"nullable": false, "required": true, "schema": {"pattern": "\\REL-[0-9a-f]{20}$", "type": "string"}}, "to_node_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}
GroupedMetricResult
{"additional_properties": false, "fields": {"group": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "value": {"nullable": false, "required": false, "schema": {"type": "union", "variants": [{"type": "integer"}, {"type": "number"}, {"type": "null"}]}}, "type": "object"}
HTTPValidationError
{"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object"}}
```

HealthResponse

```
{"additional_properties": false, "fields": {"status": {"nullable": false, "required": true, "schema": {"enum": ["UP"], "type": "string"}}}, "type": "object"}
```

ImpactScopeBlock

```
{"additional_properties": false, "fields": {"check_required": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"kind": {"nullable": false, "required": true, "schema": {"enum": ["CHAMBER", "LOT", "PARAMETER", "PROCESS_STEP", "SIBLING_CHAMBER", "WAFER"], "type": "string"}}, "relation": {"nullable": true, "required": false, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}}, "type": "object"}, "type": "array"}}, "direct": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"kind": {"nullable": false, "required": true, "schema": {"enum": ["CHAMBER", "LOT", "PARAMETER", "PROCESS_STEP", "SIBLING_CHAMBER", "WAFER"], "type": "string"}}, "relation": {"nullable": true, "required": false, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "graph_conflict": {"nullable": false, "required": false, "schema": {"default": false, "type": "boolean"}}, "reason_code": {"nullable": true, "required": false, "schema": {"pattern": "^[A-Z0-9_]+$", "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["AVAILABLE", "EMPTY"], "type": "string"}}, "summary": {"nullable": true, "required": false, "schema": {"max_length": 2000, "type": "string"}}, "type": "object"}
```

ImpactScopeItem

```
{"additional_properties": false, "fields": {"kind": {"nullable": false, "required": true, "schema": {"enum": ["CHAMBER", "LOT", "PARAMETER", "PROCESS_STEP", "SIBLING_CHAMBER", "WAFER"], "type": "string"}}, "relation": {"nullable": true, "required": false, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}
```

MetricPlan

```
{"additional_properties": false, "fields": {"column": {"nullable": true, "required": false, "schema": {"type": "string"}}, "p": {"nullable": true, "required": false, "schema": {"maximum": 100.0, "minimum": 0.0, "type": "number"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["count", "max", "mean", "median", "min", "percentile", "ratio", "std", "sum"], "type": "string"}}, "type": "object"}
```

MetrologyAskEvidence

```
{"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}
```

MetrologyEvidence

```
{"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["METROLOGY"], "type": "string"}}, "rules": ["metrology.alarm_result is forbidden"], "type": "object"}
```

NIQueryHistoryResponse

```
{"additional_properties": false, "fields": {"items": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"asked_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "error_msg": {"nullable": true, "required": false, "schema": {"type": "string"}}, "generated_sql": {"nullable": true, "required": false, "schema": {"type": "string"}}, "is_rejected": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "is_valid": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "latency_ms": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}}, "nl_query_log_id": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "outcome": {"nullable": false, "required": true, "schema": {"enum": ["DB_ERROR", "POLICY_REJECTED", "SUCCESS", "VALIDATION_FAILED"], "type": "string"}}, "question": {"nullable": false, "required": true, "schema": {"type": "string"}}, "reject_reason": {"nullable": true, "required": false, "schema": {"type": "string"}}, "row_cnt": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object"}, "type": "array"}}, "page": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "size": {"nullable": false, "required": true, "schema": {"maximum": 100.0, "minimum": 1.0, "type": "integer"}}, "total": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object"}
```

NIQueryLogItem

```
{"additional_properties": false, "fields": {"asked_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "error_msg": {"nullable": true, "required": false, "schema": {"type": "string"}}, "generated_sql": {"nullable": true, "required": false, "schema": {"type": "string"}}, "is_rejected": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "is_valid": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "latency_ms": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}}, "nl_query_log_id": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "outcome": {"nullable": false, "required": true, "schema": {"enum": ["DB_ERROR", "POLICY_REJECTED", "SUCCESS", "VALIDATION_FAILED"], "type": "string"}}, "question": {"nullable": false, "required": true, "schema": {"type": "string"}}, "reject_reason": {"nullable": true, "required": false, "schema": {"type": "string"}}, "row_cnt": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object"}
```

NIQueryOutcome

```
{"enum": ["DB_ERROR", "POLICY_REJECTED", "SUCCESS", "VALIDATION_FAILED"], "type": "string"}
```

OriginAssessment

```
{"additional_properties": false, "fields": {"basis": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"id": {"nullable": false, "required": true, "schema": {"
```



```
"max_length": 160, "min_length": 1, "type": "string"}}, "namespace": {"nullable": false, "required": true
, "schema": {"enum": ["ALARM", "CHUNK", "LOT_HIST", "PARAMETER", "RELATION"], "type": "string"}}, "type"
: "object"}, "type": "array"}}, "compared": {"nullable": false, "required": true, "schema": {"additional_
properties": false, "fields": {"downstream": {"nullable": false, "required": false, "schema": {"default":
"NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "history": {"n
ullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAIL
ABLE", "NOT_CHECKED"], "type": "string"}}, "metrology": {"nullable": false, "required": false, "schema":
{"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "si
bling": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED",
"NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "upstream": {"nullable": false, "required": false,
"schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "stri
ng"}}, "type": "object"}}, "scope": {"nullable": false, "required": true, "schema": {"enum": ["CURRENT_C
HAMBER", "DOWNSTREAM", "EQUIPMENT_COMMON", "UNDETERMINED", "UPSTREAM"], "type": "string"}}, "type": "obj
ect"}
```

OriginBasisRef

```
{"additional_properties": false, "fields": {"id": {"nullable": false, "required": true, "schema": {"max_l
ength": 160, "min_length": 1, "type": "string"}}, "namespace": {"nullable": false, "required": true, "sch
ema": {"enum": ["ALARM", "CHUNK", "LOT_HIST", "PARAMETER", "RELATION"], "type": "string"}}, "type": "obj
ect"}
```

ParameterFinding

```
{"additional_properties": false, "fields": {"direction": {"nullable": false, "required": true, "schema":
{"enum": ["ABOVE", "BELOW", "BOTH"], "type": "string"}}, "excursion_ratio": {"nullable": false, "required
": true, "schema": {"type": "number"}}, "lot_hist_ids": {"nullable": false, "required": true, "schema": {
"items": {"type": "string"}, "type": "array"}}, "parameter_id": {"nullable": false, "required": true, "sc
hema": {"max_length": 20, "min_length": 1, "type": "string"}}, "step_no": {"nullable": false, "required":
true, "schema": {"minimum": 1.0, "type": "integer"}}, "wafer_scope": {"nullable": false, "required": tru
e, "schema": {"enum": ["ALL", "PARTIAL", "SINGLE"], "type": "string"}}, "type": "object"}
```

ParameterItem

```
{"additional_properties": false, "fields": {"LCL": {"nullable": true, "required": true, "schema": {"type"
: "number"}}, "LSL": {"nullable": true, "required": true, "schema": {"type": "number"}}, "TARGET": {"null
able": false, "required": true, "schema": {"type": "number"}}, "UCL": {"nullable": false, "required": tru
e, "schema": {"type": "number"}}, "USL": {"nullable": false, "required": true, "schema": {"type": "number
"}}, "area": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"
}}, "ctrl_lower": {"nullable": true, "required": true, "schema": {"type": "number"}}, "ctrl_upper": {"nul
lable": false, "required": true, "schema": {"type": "number"}}, "name": {"nullable": false, "required": t
rue, "schema": {"min_length": 1, "type": "string"}}, "parameter_id": {"nullable": false, "required": true
, "schema": {"min_length": 1, "type": "string"}}, "parameter_name": {"nullable": false, "required": true,
"schema": {"min_length": 1, "type": "string"}}, "spec_lower": {"nullable": true, "required": true, "sche
ma": {"type": "number"}}, "spec_upper": {"nullable": false, "required": true, "schema": {"type": "number"
}}, "target_value": {"nullable": false, "required": true, "schema": {"type": "number"}}, "unit": {"nullab
le": true, "required": true, "schema": {"type": "string"}}, "upper_only": {"nullable": false, "required":
true, "schema": {"type": "boolean"}}, "type": "object"}
```

PostActionObservationBlock

```
{"additional_properties": false, "fields": {"message": {"nullable": false, "required": false, "schema": {
"default": "\u00d5\u00885 \u00815\u00801 \u00370\u00774\u00d130\u0014b\u005d0\u00b294 \u00870\u00ce58 \u00774\u00d6c4 \u00ac
f5\u00815 \u00ad00\u00ce21\u00ac12\u00774 \u005c6\u005b4 \u00d6a8\u00acfc\u00b97c \u00d3c9\u00ac00\u00d560 \u00218 \u005c6\u00c74c"
, "type": "string"}}, "status": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILAB
LE_STATIC_DATASET", "type": "string"}}, "type": "object"}
```

PublicAgentRunItem

```
{"additional_properties": false, "fields": {"action_id": {"nullable": true, "required": true, "schema": {
"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"mi
n_length": 1, "type": "string"}}, "alarm_id": {"nullable": false, "required": true, "schema": {"min_lengt
h": 1, "type": "string"}}, "alarm_source": {"nullable": false, "required": true, "schema": {"enum": ["R03
", "SUMMARY", "TRACE"], "type": "string"}}, "approval_id": {"nullable": true, "required": true, "schema":
{"min_length": 1, "type": "string"}}, "chamber": {"nullable": false, "required": true, "schema": {"min_l
ength": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_lengt
h": 1, "type": "string"}}, "confidence": {"nullable": true, "required": true, "schema": {"maximum": 1.0, "
minimum": 0.0, "type": "number"}}, "created_at": {"nullable": false, "required": true, "schema": {"format
": "date-time", "type": "string"}}, "deliveries": {"nullable": false, "required": true, "schema": {"items
": {"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema":
{"enum": ["EMAIL", "MES"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema":
{"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}},
"type": "object"}, "type": "array"}}, "fault_code": {"nullable": true, "required": true, "schema": {"en
um": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "fault_color": {"nullable": false, "require
d": true, "schema": {"type": "null"}}, "fault_name": {"nullable": false, "required": true, "schema": {"ty
pe": "null"}}, "latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "in
teger"}}, "llm_model": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"
}}, "predicted_fault_code": {"nullable": true, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH",
"RFM", "TMD"], "type": "string"}}, "recommended_action": {"nullable": true, "required": true, "schema":
{"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "status": {"nullable": false, "requir
ed": true, "schema": {"enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}},
"tools": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fie
lds": {"n": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result
_summary": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "s": {"n
```

```

    "nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}},
    "status": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}},
    "tool_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}},
    "type": "object"},
    "type": "array"}},
    "type": "object"}
PublicApprovalDecision
{"enum": ["APPROVED", "REJECTED"], "type": "string"}
PublicApprovalItem
{"additional_properties": false, "fields": {"action_code": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approved_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "approved_by": {"nullable": true, "required": true, "schema": {"type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_by": {"nullable": true, "required": true, "schema": {"type": "string"}}, "decision_comment": {"nullable": true, "required": true, "schema": {"type": "string"}}, "equipment": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "lot": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "predicted_fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "reason": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "type": "object"}
PublicApprovalStatus
{"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}
PublicDeliveryChannel
{"enum": ["EMAIL", "MES"], "type": "string"}
PublicDeliveryItem
{"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, "type": "object"}
PublicToolCallItem
{"additional_properties": false, "fields": {"n": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result_summary": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "s": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "tool_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}
ReactStepPublic
{"additional_properties": false, "fields": {"argument_summary": {"nullable": true, "required": false, "schema": {"pattern": "^\\ud6c4\\ubcf4 wafer [1-9][0-9]*\\((\\ud604\\uc7ac\\uc0c1\\ub958\\ud558\\ub958)\\)\\uc774\\ub825 [1-9][0-9]*\\((\\ud604\\uc7ac\\ud615\\uc81c)\\)\\uacc4\\uce21 [1-9][0-9]*\\((\\ud604\\uc7ac\\uc0c1\\ub958\\ud558\\ub958)\\)\\ubb38\\uc11c \\uc80\\uc0c9\\uc124\\ube44 \\ucee8\\ud14d\\uc2a4\\ud2b8)$", "type": "string"}}, "degraded": {"nullable": false, "required": false, "schema": {"default": false, "type": "boolean"}}, "guard_code": {"nullable": true, "required": false, "schema": {"max_length": 64, "type": "string"}}, "observation_summary": {"nullable": true, "required": false, "schema": {"max_length": 160, "type": "string"}}, "phase": {"nullable": false, "required": true, "schema": {"enum": ["OBSERVED", "REJECTED", "SELECTED", "STOPPED"], "type": "string"}}, "rationale_summary": {"nullable": true, "required": false, "schema": {"max_length": 120, "type": "string"}}, "react_prompt_version": {"nullable": true, "required": false, "schema": {"type": "string"}}, "selector_tokens": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"input": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "output": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object"}}, "seq": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "stop_reason": {"nullable": true, "required": false, "schema": {"enum": ["BUDGET_EXHAUSTED", "GUARD_LIMIT", "LLM_DEPENDENCY", "LLM_STOP", "LLM_TIMEOUT", "REACT_STRUCTURE_INVALID", "STEP_CAP"], "type": "string"}}, "tool": {"nullable": true, "required": false, "schema": {"enum": ["get_chamber_parameter_history", "get_equipment_context", "get_fdc_summary", "get_metrology_result", "search_documents", "stop"], "type": "string"}}, "type": "object"}
ReadinessCheck
{"additional_properties": false, "fields": {"latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0, "type": "integer"}}, "reason_code": {"nullable": true, "required": true, "schema": {"enum": ["CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["FAIL", "PASS"], "type": "string"}}, "type": "object"}
ReadinessChecks
{"additional_properties": false, "fields": {"kafka": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0, "type": "integer"}}, "reason_code": {"nullable": true, "required": true, "schema": {"enum": [
```

```

CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY",
"TIMEOUT"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["FA
IL", "PASS"], "type": "string"}}, "type": "object"}}, "n8n": {"nullable": false, "required": true, "sche
ma": {"additional_properties": false, "fields": {"latency_ms": {"nullable": false, "required": true, "sch
ema": {"minimum": 0, "type": "integer"}}, "reason_code": {"nullable": true, "required": true, "schema": {
"enum": ["CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_N
OT_READY", "TIMEOUT"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"en
um": ["FAIL", "PASS"], "type": "string"}}, "type": "object"}}, "neo4j": {"nullable": false, "required":
true, "schema": {"additional_properties": false, "fields": {"latency_ms": {"nullable": false, "required":
true, "schema": {"minimum": 0, "type": "integer"}}, "reason_code": {"nullable": true, "required": true,
"schema": {"enum": ["CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "
RAG_MODEL_NOT_READY", "TIMEOUT"], "type": "string"}}, "status": {"nullable": false, "required": true, "sc
hema": {"enum": ["FAIL", "PASS"], "type": "string"}}, "type": "object"}}, "postgresql_runtime": {"nullab
le": false, "required": true, "schema": {"additional_properties": false, "fields": {"latency_ms": {"nulla
ble": false, "required": true, "schema": {"minimum": 0, "type": "integer"}}, "reason_code": {"nullable":
true, "required": true, "schema": {"enum": ["CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STA
LE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"], "type": "string"}}, "status": {"nullable": fals
e, "required": true, "schema": {"enum": ["FAIL", "PASS"], "type": "string"}}, "type": "object"}}, "rag":
{"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"latency_ms"
: {"nullable": false, "required": true, "schema": {"minimum": 0, "type": "integer"}}, "reason_code": {"nu
llable": true, "required": true, "schema": {"enum": ["CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFK
A_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"], "type": "string"}}, "status": {"nullab
le": false, "required": true, "schema": {"enum": ["FAIL", "PASS"], "type": "string"}}, "type": "object"}
}, "reference_migration": {"nullable": false, "required": true, "schema": {"additional_properties": false
, "fields": {"latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0, "type": "intege
r"}}, "reason_code": {"nullable": true, "required": true, "schema": {"enum": ["CONTRACT_MISMATCH", "DEPEN
DENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"], "type": "stri
ng"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["FAIL", "PASS"], "type": "stri
ng"}}, "type": "object"}}, "type": "object"}
ReadinessResponse
{"additional_properties": false, "fields": {"checks": {"nullable": false, "required": true, "schema": {"a
dditional_properties": false, "fields": {"kafka": {"nullable": false, "required": true, "schema": {"addit
ional_properties": false, "fields": {"latency_ms": {"nullable": false, "required": true, "schema": {"mini
mum": 0, "type": "integer"}}, "reason_code": {"nullable": true, "required": true, "schema": {"enum": ["CO
NTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "
TIMEOUT"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["FAIL",
"PASS"], "type": "string"}}, "type": "object"}}, "n8n": {"nullable": false, "required": true, "schema"
:"additional_properties": false, "fields": {"latency_ms": {"nullable": false, "required": true, "schema"
:"minimum": 0, "type": "integer"}}, "reason_code": {"nullable": true, "required": true, "schema": {"en
um": ["CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_
READY", "TIMEOUT"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum"
: ["FAIL", "PASS"], "type": "string"}}, "type": "object"}}, "neo4j": {"nullable": false, "required": tru
e, "schema": {"additional_properties": false, "fields": {"latency_ms": {"nullable": false, "required": tr
ue, "schema": {"minimum": 0, "type": "integer"}}, "reason_code": {"nullable": true, "required": true, "sc
hema": {"enum": ["CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG
_MODEL_NOT_READY", "TIMEOUT"], "type": "string"}}, "status": {"nullable": false, "required": true, "schem
a": {"enum": ["FAIL", "PASS"], "type": "string"}}, "type": "object"}}, "postgresql_runtime": {"nullable"
: false, "required": true, "schema": {"additional_properties": false, "fields": {"latency_ms": {"nullabl
e": false, "required": true, "schema": {"minimum": 0, "type": "integer"}}, "reason_code": {"nullable": tru
e, "required": true, "schema": {"enum": ["CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE",
"NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"], "type": "string"}}, "status": {"nullable": false,
"required": true, "schema": {"enum": ["FAIL", "PASS"], "type": "string"}}, "type": "object"}}, "rag": {"
nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"latency_ms": {
"nullable": false, "required": true, "schema": {"minimum": 0, "type": "integer"}}, "reason_code": {"nulla
ble": true, "required": true, "schema": {"enum": ["CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_L
AG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"], "type": "string"}}, "status": {"nullable"
: false, "required": true, "schema": {"enum": ["FAIL", "PASS"], "type": "string"}}, "type": "object"}},
"reference_migration": {"nullable": false, "required": true, "schema": {"additional_properties": false, "
fields": {"latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0, "type": "integer"}
}, "reason_code": {"nullable": true, "required": true, "schema": {"enum": ["CONTRACT_MISMATCH", "DEPENDE
CY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"], "type": "string"
}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["FAIL", "PASS"], "type": "string"
}}, "type": "object"}}, "type": "object"}}, "dataset_epoch": {"nullable": false, "required": true, "sch
ema": {"enum": ["fdc_final_20260818"], "type": "string"}}, "status": {"nullable": false, "required": true
, "schema": {"enum": ["NOT_READY", "READY"], "type": "string"}}, "rules": ["all checks PASS yields statu
s READY and HTTP 200", "any check FAIL yields status NOT_READY and HTTP 503"], "type": "object"}
RunAlarmEvidence
{"additional_properties": false, "fields": {"alarm": {"nullable": false, "required": true, "schema": {"ad
ditional_properties": false, "fields": {"alarm_id": {"nullable": false, "required": true, "schema": {"min
_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03",
"SUMMARY", "TRACE"], "type": "string"}}, "type": "object"}}, "excerpt": {"nullable": false, "required":
true, "schema": {"min_length": 1, "type": "string"}}, "source_id": {"nullable": false, "required": tru
e, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema"
: {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type":

```

```
"string"}}, "type": "object"}
RunStatus
{"enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}
SelectorTokens
{"additional_properties": false, "fields": {"input": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "output": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}}, "type": "object"}
SimilarIncidentItem
{"additional_properties": false, "fields": {"agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "parameter_jaccard": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "predicted_fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "recommended_action": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "score": {"nullable": false, "required": true, "schema": {"maximum": 100.0, "minimum": 0.0, "type": "integer"}}}, "type": "object"}
SimilarIncidentsBlock
{"additional_properties": false, "fields": {"items": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "parameter_jaccard": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "predicted_fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "recommended_action": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "score": {"nullable": false, "required": true, "schema": {"maximum": 100.0, "minimum": 0.0, "type": "integer"}}}, "type": "object"}, "type": "array"}, "label": {"nullable": false, "required": false, "schema": {"default": "\uace0\u815 \u2dc\u5f0 \ub370\u2774\u2d130 \ub0b4 \ube44\u2d50 \uacbd\uacfc", "type": "string"}}, "reason_code": {"nullable": true, "required": false, "schema": {"pattern": "[A-Z0-9_]+$", "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["AVAILABLE", "EMPTY"], "type": "string"}}}, "type": "object"}
SqlValidateRequest
{"additional_properties": false, "fields": {"sql": {"nullable": false, "required": true, "schema": {"max_length": 20000, "min_length": 1, "type": "string"}}, "type": "object"}
SqlValidateResponse
{"additional_properties": false, "fields": {"checks": {"nullable": true, "required": false, "schema": {"items": {"additional_properties": false, "fields": {"key": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "label": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "ok": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "type": "object"}, "type": "array"}, "normalized_sql": {"nullable": true, "required": false, "schema": {"type": "string"}}, "reason": {"nullable": false, "required": true, "schema": {"type": "string"}}, "valid": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "type": "object"}
ToolCallStatus
{"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}
TraceAskEvidence
{"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}
TraceEvidence
{"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["TRACE"], "type": "string"}}, "type": "object"}
TracePoint
{"additional_properties": false, "fields": {"measured_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "recipe_step_no": {"nullable": false, "required": true, "schema": {"minimum": 1, "type": "integer"}}, "seq_no": {"nullable": false, "required": true, "schema": {"minimum": 0, "type": "integer"}}, "value": {"nullable": false, "required": true, "schema": {"type": "number"}}, "type": "object"}
ValidationCheck
{"additional_properties": false, "fields": {"key": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "label": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "ok": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "type": "object"}
ValidationError
{"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}
VisualizationPlan
```

```
{"additional_properties": false, "fields": {"chart_type": {"nullable": false, "required": true, "schema": {"enum": ["bar", "histogram", "line", "table"], "type": "string"}}, "x": {"nullable": true, "required": false, "schema": {"type": "string"}}, "y": {"nullable": true, "required": false, "schema": {"type": "string"}}}, "type": "object"}
AUDIT EVENTS
DETECTION_COMPLETED / LOT_HIST / A
AGENT_RUN_STARTED / AGENT_RUN / C
HYPOTHESIS_GENERATED / AGENT_RUN / C
APPROVAL_REQUESTED / APPROVAL / C
APPROVAL_DECIDED / APPROVAL / C
ACTION_SENT / ACTION / C delivery service
ACTION_SEND_FAILED / ACTION / C delivery service
AGENT_RUN_COMPLETED / AGENT_RUN / C
AGENT_RUN_FAILED / AGENT_RUN / C
```